

ENG511
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COGNITIVE PROCESSES

Topic-133: Cognitive Processes

We saw that parents provide a structured environment for children who are acquiring language. Although some of these speech adaptations facilitate development, they are not sufficient to explain language acquisition. To benefit from these language lessons, children must have certain cognitive prerequisites. These include procedures for registering, storing, and analyzing linguistic information. A simple analogy may be helpful here. Suppose you are taking a course in philosophy. The instructor is well prepared, lectures well, and is available when students have problems. Although all of these characteristics are beneficial, they do not guarantee the desired learning outcome. A course in philosophy typically requires students to think abstractly and write analytical essays. Students who lack these skills may have considerable difficulty even if the course material is presented in an organized fashion. The same is true for the child learning language. A structured environment is helpful only if the child has the ability to take advantage of the structure that is provided.

Topic-134: Operating Principles

One of the most productive approaches to the question has been Slobin's work on operating principles. We may think of operating principles as children's preferred ways of taking in (or operating on) information. These principles have proven useful in explaining certain patterns in early child grammar. For instance, children in virtually all languages use fixed word order to create meanings, even though some languages have much freer word order than others.

One problem with the notion of operating principles is that it is open to the charge of circularity. The evidence for operating principles is found in children's language patterns, and then the principles are assumed to account for the patterns. Independent evidence of these operating principles would be helpful. It might be possible to see evidence of these principles in cognitive domains other than language, but, as written, they tend to be fairly specific to language. Slobin is noncommittal on the question of whether these processes are specific to language (modular) or whether they can be understood in terms of general cognitive processes. We turn now to cognitive processes that are more clearly independent of language.

Topic-135: Sensorimotor Schemata

Another way of characterizing the child's cognitive system comes from the work of Piaget who expresses the belief that children undergo several qualitative shifts in their thinking throughout development. Piaget refers to the first 2 years as the sensorimotor period of development because the schemata the child uses to organize experience are directly related to taking in sensory information and acting on it. Sensorimotor schemata include banging, sucking, and throwing. The major development that culminates near the end of the sensorimotor period is the acquisition of object permanence, the notion that objects continue to exist even when they cannot be perceived. Once object permanence is acquired, the child is no longer at the mercy of immediate stimuli but can respond on the basis of stimuli no longer present. We would certainly anticipate that developments of this magnitude would be related to the child's language development. More specifically, we can make two predictions about child language. One is that the very young infant, who has not yet acquired object permanence, should use words that refer to

concrete objects in the immediate environment, especially those that the child easily manipulates. This appears to be the case because early child language consists of a large number of “here and now” words.

Topic-136: Whole Object Bias and Taxonomic Bias

Clearly adults present children with a simplified and orderly pattern of data that would seem to facilitate vocabulary development. The question to be pursued here is not whether such adaptations are necessary for normal language acquisition but whether they are sufficient. As children are exposed to adult words for objects, many referents are possible for these words. It seems unlikely that children explore every possible meaning of a given word, given what we have learned about the speed of lexical acquisition. Adult naming practices help the child, but some theorists believe that the child must have certain expectations about word learning to benefit maximally from these lessons. The notion of a cognitive constraint is that children are constrained to consider only some of these possibilities or at least to give priority to them over others. A subsequent study presented children with novel labels for objects for which they already had labels, such as claw for hammer. In this instance, the children interpreted the novel term as applying to only one part of the object.

Research on cognitive prerequisites for language development has proceeded on the assumption that certain cognitive processes must be in place for the child to benefit from structured language lessons. We have considered two types of cognitive processes that may assist or guide language development. Operating principles are preferred ways of taking in linguistic information. Sensorimotor schemata are ways of organizing the world that emerge in the first two years of life. The general prediction that the cognitive position makes is that children with a given cognitive prerequisite should acquire corresponding aspects of language more rapidly than those without the prerequisite. Studies of sensorimotor schemata, however, suggest that cognitive processes do not emerge prior to language but rather simultaneously with language. In addition, we have discussed the role of cognitive constraints in children’s vocabulary acquisition. Considerable evidence suggests that constraints guide children’s acquisition of lexical items. There is continued debate regarding the best way to characterize these constraints. Finally, we saw that in certain individuals there are dissociations between language and cognition: relatively strong cognitive skills with weak linguistic skills or relatively strong linguistic skills with weak cognitive skills. These observations suggest that cognitive development, although it is generally associated with language development, may not be either necessary or sufficient for it.

When children encounter a new label, they prefer to attach the label to the entire object rather than to part of the object. To return to the earlier example, when someone points to an object and says dog, the child assumes that the word is a label for the entire object rather than the dog’s tail.

Whole object bias in developmental psychology: For example, if a child is shown and given the label "truck", the child will assume truck refers to the whole object instead of the tires, doors, colors or other parts. If a researcher points to an object while simultaneously saying a new name, children will assume that the new label refers to the whole object. Ellen Markman pioneered work in this field. Her studies suggest that even in cases where color or dynamic activities are made salient to children, they will still interpret the new word as a label for whole objects. Furthermore, infants hold a primitive theory of the physical world that is guided by three constraints on the behavior of physical bodies: objects must move as wholes, objects move independently of each other, and objects move on connected paths. It is

suggested that these three constraints help guide children's interpretations of scenes, and, in turn, explain how the whole object bias reflects the nonlinguistic status of objects.

In addition, children seem to use a taxonomic bias: they will assume that the object label is a taxonomic category rather than a name for an individual dog. For example, they will assume that dog is a label for a group of animals, not just Fido. Ordinarily, children focus on thematic relations between objects when categorizing. If given milk, a spoon, or a car, children will group each item with a cow, soup, or a stop sign, respectively. However, when children hear a new label they shift their attention to taxonomic relationships, even though they consider thematic relations to be a good way of organizing the objects themselves. Instead of cow being linked to milk, it would be linked to pig or horse. The new word is assumed to refer to other objects within the same taxonomic category. Ellen Markman's early work showed this constraint at work. When two- and three-year-olds were presented with two basic-level objects, two different kinds of dogs, and a third thematically related object, dog food, they showed a tendency to select a dog and dog food; however, if one of the dogs was labeled with an unfamiliar word, the children were more likely to select the two dogs.

Topic-137: Mutual Exclusivity Bias

It refers to the fact that a child who knows the name of a particular object will then generally reject applying a second name to that object. Several experimental studies by Markman and Wachtel have supported the notion that children use mutual exclusivity in acquiring new words. In one, 3-year-old children were presented with pairs of objects. One member of each pair was an object for which the child already had a label (such as a banana, toy cow, or spoon), and the other was an object for which the child did not have a label (such as a lemon wedge press or a pair of tongs). The children were then asked by a puppet to show me the x (x was a nonsense syllable). The children were much more likely to select the novel object. A subsequent study presented children with novel labels for objects for which they already had labels, such as claw for hammer. In this instance, the children interpreted the novel term as applying to only one part of the object.

Topic-138: Impairments of Language and Cognition

Our knowledge of the relationships between language and cognition has also been advanced by studies of children and adolescents with cognitive or linguistic impairments. The notion that a close relationship exists between language and cognition has generally been supported by studies of individuals with Down syndrome (reviewed by Rosenberg, 1982; but see Rondal, 1993). These individuals tend to have language delays that are proportionate to the severity of their cognitive disability. However, in certain individuals, there can be significant discrepancies between the level of cognitive functioning and the level of linguistic functioning.

INNATE MECHANISM

Topic-139: The Language Bioprogram Hypothesis

One version of how innate processes operate in child language has been called the language bioprogram hypothesis by Bickerton (1981, 1983, 1984, 1999). Bickerton's claim, is that we, as children, have an innate grammar that is available biologically if our language input is insufficient to acquire the language of our community. It is something like a linguistic backup system.

Pidgins and Creoles: To understand this idea more fully, we have to make a few distinctions. A pidgin is 'an auxiliary language that arises when speakers of several mutually unintelligible languages are in close contact' (Bickerton, 1984, p. 173). Typically this occurs when workers from diverse countries are brought in as cheap labor in an agricultural community. Immigrant workers come to speak a simpler form of the dominant language of the area—just enough to get by. A creole occurs when the children of these immigrants acquire a pidgin as their native language. Because access to native speakers of the dominant language is usually limited, these children receive the impoverished pidgin version as their primary linguistic input. Bickerton (1983) observes that the conditions necessary to produce creoles have existed numerous times between 1500 and 1900 when various European nations developed labor-intensive agricultural economies on isolated, under populated tropical islands throughout the world. Bickerton's studies have focused on creoles in Hawaii. Although Hawaiian contact with Europeans goes back to the 18th century, it was not until 1876 that a revision of the U.S. tariff laws led to a large influx of indentured workers to harvest Hawaiian sugar. Because Hawaiian creole developed between 1900 and 1920, it was possible for Bickerton to study the development of the creole by studying the speech of people who are still living. In particular, he examined the language of immigrants who moved to Hawaii and that of their children who were born in the first two decades of the 20th century. The speech of pidgin speakers was rudimentary. In many cases, there was no recognizable syntax, and the language resembled a linguistic free-for-all. Some speakers used one word order and others another; the word orders were often related to the speaker's own native language. Moreover, complex sentences were absent in pidgin: pidgin sentences had no subordinate clauses, and even single-clause utterances often lacked verbs. In addition, there was no consistent system of anaphora.

Topic-140: The Language Bioprogram

Despite this impoverished linguistic input, the children of immigrants developed a creole that was relatively sophisticated (Roberts, 1998). It included consistent word order, the use of complex sentences with relative clauses, and the distinction between definite and indefinite articles. Unlike pidgins, the creoles resembled the structural rules of other languages. From these observations, Bickerton concludes that children have an innate grammar that, in the absence of proper environmental input, serves as the child's language system. He calls this system the language bioprogram. Bickerton (1984) has responded to other possible interpretations of his research. One is that the sophistication found in the children's creoles was based on their access to English, the language of the plantation owners. Bickerton points out, however, that contact between immigrant families and owners was limited and that the Hawaiian creole differed in several respects from English. Another possibility is that linguistic features not attributable to English could be derived from the original native languages of the parents. For example, children whose

parents were Portuguese might incorporate some Portuguese elements into their creoles. Again, the evidence provided by Bickerton suggests otherwise; he found that Hawaiian creole was strikingly similar to creoles created by children in very different parts of the world. The language bio program hypothesis has been further buttressed by studies of language development in congenitally deaf children by Goldin-Meadow and her colleagues. Task specificity refers to the notion that the cognitive processes associated with language use are not general-purpose problem-solving processes but are instead restricted to language. Bickerton goes a step further: Not only is the language bio program specific to language, but it is itself highly specific—a prepackaged, ready-to-go linguistic system.

Topic-141: Parameter Setting

Parameters is a framework within generative linguistics in which the syntax of a natural language is described in accordance with general principles (i.e. abstract rules or grammars) and specific parameters (i.e. markers, switches) that for particular languages are either turned on or off. For example, the position of heads in phrases is determined by a parameter.

Bickerton's language bio program may be thought of as a specific instance of a general innate mechanism called parameter setting. The notion of parameters plays a key role in the concept of universal grammar (Chomsky, 1981). In this view, grammar can be defined in terms of a set of parameters corresponding to each of the subsystems of the language, with each parameter having a finite (usually small) number of possible settings. Various combinations of these parameter settings then yield all of the languages of the world. According to Chomsky (1981), children are born with the knowledge of the parameters and their possible settings. The task of acquiring a language is therefore reduced to identifying which parameter settings apply to one's native language. One parameter is called the head parameter and has been discussed by Cook (1988). Each phrase in the language has one element that is most essential, which is called the head. It is the noun in noun phrases and the verb in verb phrases. The head parameter specifies the position of the head within the phrase.

Topic-142: The Subset Principle

Another way to think about how these parameter settings may be made is through what Berwick and Weinberg (1984) call the subset principle. First, think about languages as subsets of one another. Consider word order. English is a very strict word-order language; Russian allows a small set of admissible orders, and the aborigine language Warlpiri allows an almost total scrambling of word order within a clause. The idea of a subset is that Russian could be considered a subset of Warlpiri with somewhat more restricted word order. In the same way, with respect to word order, English may be considered a subset of Russian.

Topic-143: The Issues of Negative Evidence

One important feature of the way in which the subset principle was formulated deals with the distinction between positive and negative evidence. At the grammatical level, positive evidence is evidence that a particular utterance is grammatical in the language that the child is learning; negative evidence is evidence that a particular utterance is ungrammatical. Children receive positive evidence when they are exposed to an utterance that is not corrected or otherwise indicated as inappropriate.

Children receive negative evidence when someone indicates that a particular utterance is ungrammatical or inappropriate. Pinker (1990) argues that it would be very difficult to acquire a language from positive evidence alone. This notion is based, in part, on some computer simulation studies of language learning done some time ago. This work assumed that children use linguistic evidence to construct hypotheses about the language they are learning, much as a linguist would use such evidence to learn about a language in a foreign land. Gold found that when he wrote a program in which the computer received only positive evidence, it failed to acquire the language adequately. This was presumably because positive evidence is consistent with a great number of different grammars. Without knowing what is ungrammatical in a language, it is impossible to rule out some of the various competing grammars. Pinker (1990) has claimed that, on the whole, parents do not provide sufficient negative evidence to enable a child to learn a language. He argues that although negative evidence is sometimes present, it is not systematically and consistently available to all children acquiring a language, and yet all normal children do acquire a language. Therefore, innate linguistic mechanisms, such as the subset principle, are needed to constrain the child's search processes. Pinker's argument is as follows: A: Positive evidence alone is consistent with too many competing grammars. B: Negative evidence, which could constrain the problem space, is not generally available. C: Therefore, some constraints must be innate. We know that children are not always receptive to the adults' observations about their grammar.

Topic-144: Objections to Innate Mechanisms

To sum up this section so far, studies of pidgins and creoles suggest the presence of an innate backup grammar, the language bio program. Researchers studying parameters have attempted to specify what kinds of linguistic information must be innately present before children can take advantage of the language they receive from their environment. Studies of negative evidence suggest that such evidence is not pervasive enough to present a full account of language acquisition. All of these studies converge on the conclusion that some innate linguistic mechanisms—in conjunction with environmental and cognitive factors—must be present in order for children to acquire language as successfully as most children do. This conclusion has been challenged by Elman, Bates, Johnson, Karmiloff-Smith, Parisi, and Plunkett (1996), who raise a number of objections to innate language mechanisms. Let us consider some of these objections and what we can say about them. Discussing the studies of Nicaraguan Sign Language presented earlier in the chapter, Elman and colleagues make the following point:

We would agree that these phenomena are extremely interesting, and that they attest to a robust drive among human beings to communicate their thoughts as rapidly and efficiently as possible. However, these phenomena do not require a preformationist scenario (i.e., a situation in which the grammar emerges because it was innately specified). If children develop a robust drive to solve this problem, and are born with processing tools to solve it, then the rest may simply follow because it is the natural solution. (p. 390)

Jackendoff has responded that adults also have a 'robust drive' to acquire language, but they acquire pidgins rather than creoles. Thus, something beyond a 'robust drive' seems to be necessary. According to Jackendoff and others that something is an innate mechanism for acquiring language.

BRIAN MECHANISM AND LANGUAGE

Topic-145: Broca's Aphasia

A person with expressive aphasia will exhibit effortful speech. Speech generally includes important content words, but leaves out function words that have only grammatical significance and not real-world meaning, such as prepositions and articles. This is known as "telegraphic speech." The person's intended message may still be understood but his or her sentence will not be grammatically correct. In very severe forms of expressive aphasia, a person may only speak using single word utterances. Typically, comprehension is mildly to moderately impaired in expressive aphasia due to difficulty in understanding complex grammar.

The speech of a person with expressive aphasia contains mostly content words such as nouns, verbs, and some adjectives. However, function words like conjunctions, articles, and prepositions are rarely used except for "and" which is prevalent in the speech of most patients with aphasia. The omission of function words makes the person's speech agrammatic. A communication partner of a person with aphasia may say that the person's speech sounds telegraphic due to poor sentence construction and disjointed words. For example, a person with expressive aphasia might say "Smart... university... smart... good... good..."

Topic-146: Wernicke's and Conduction Aphasia

The following are common symptoms seen in patients with Wernicke's aphasia:

Impaired comprehension: It deficits in understanding (receptive) written and spoken language.

This is because Wernicke's area is responsible for assigning meaning to the language that is heard, so if it is damaged, the brain cannot comprehend the information that is being received.

Poor word retrieval: The ability to retrieve target words is impaired. This is also referred to as Anomia.

Fluent speech: Individuals with Wernicke's aphasia do not have difficulty with producing connected speech that flows. Although the connection of the words may be appropriate, the words they are using may not belong together or make sense (see Production of Jargon below).

Production of jargon: Speech that lacks content, consists of typical intonation, and is structurally intact. Jargon can consist of a string of neologisms, as well as a combination of real words that do not make sense together in context. It may include word salads.

Awareness: Individuals with Wernicke's aphasia are often not aware of their incorrect productions which would further explain why they do not correct themselves when they produce jargon, paraphasias, or neologisms.

Conduction aphasics will show relatively well-preserved auditory comprehension, which may even be completely functional. Spontaneous speech production will be fluent and generally grammatically and syntactically correct. Intonation and articulation will also be preserved. Speech will often contain paraphasic errors: phonemes and syllables will be dropped or transposed (e.g., "snowball" → "snowall",

"television" → "vellitision", "ninety-five percent" → "ninety-twenty percent"). The hallmark deficit of this disorder, however, is in repetition. Patients will show a marked inability to repeat words or sentences when prompted by an examiner.

After saying a sentence to a person with conduction aphasia, he or she will be able to paraphrase the sentence accurately but will not be able to repeat it, possibly because their "motor speech error processing is disrupted by inaccurate forward predictions, or because detected errors are not translated into corrective commands due to damage to the auditory-motor interface."

Topic-147: Other Aphasias

One rare form of aphasia is called pure word deafness. Behaviorally, such individuals are unable to comprehend language in the auditory modality, although they are still capable of comprehending visual language and producing language in either modality. Anatomically, there is damage to the auditory nerve, which sends messages to the auditory centers in the left hemisphere. In addition, there is a loss of those portions of the corpus callosum (the thick band of fibers that connect the two hemispheres) that send messages from the auditory region in the right hemisphere to the language areas, particularly Wernicke's area, in the left hemisphere. The result is that neither the left nor the right auditory center can transmit information to the language regions, so even though patients can hear some words, they cannot comprehend them. Another interesting form of aphasia is called alexia, which is the dissociation (disconnection) of the visual regions from the language areas. In its most severe form, alexia prevents even the recognition of individual letters or matching of script and print (Goodglass & Geschwind, 1976). Damage to the angular gyrus leads to both alexia and agraphia, the inability to write. It is thought that the angular gyrus serves as an association area in the brain that connects one region with another. In particular, it is important for the association of visual stimuli with linguistic symbols, which influences both reading and writing. Alexia also sometimes occurs without agraphia; in one case (Dejerine, 1892, cited in Geschwind, 1965), damage to the visual cortex on the left side was coupled with an injury to the portion of the corpus callosum that connected an intact right visual area with the language areas on the left.

Topic-148: Geschwind's Models of Language Processing

1. For listening to and understanding spoken words, the sounds of the words are sent through the auditory pathways to the primary auditory cortex (Heschl's gyrus). From there, they continue to Wernicke's area, where the meaning of the words is extracted.
2. In order to speak, the meanings of words are sent from Wernicke's area via the arcuate fasciculus to Broca's area, where morphemes are assembled. The model proposes that Broca's area holds a representation for articulating words. Instructions for speech are sent from Broca's area to the facial area of the motor cortex, and from there instructions are sent to facial motor neurons in the brainstem, which relay movement orders to facial muscles.
3. In order to read, information concerning the written text is sent from visual areas to the angular gyrus and from there to Wernicke's area, for silent reading or, together with Broca's area, for reading out loud.

Topic-149: Experimental Studies of Aphasia

Let us now look at psycholinguistic research that has clarified the role of syntactic and semantic processes in various aphasia. The traditional view has been that Broca's or agrammatic aphasia is a production deficit and Wernicke's a comprehension deficit. The implicit assumption is that the underlying language representation is intact with Broca's patients but that they have difficulty putting appropriately formulated linguistic messages into words.

Conceptualize sentences normally but have difficulty translating them into productive speech. Clearly, data regarding the comprehension abilities of Broca's aphasics would be extremely useful here. Broca's aphasics have often been viewed as having normal comprehension, but we have good reasons for questioning this assumption. For one, many of the tests of comprehension have been extremely global in nature and do not clarify the respective roles of syntactic and semantic processes in comprehension. For another, it is possible to disguise deficits in comprehension somewhat more easily than deficits in production. The latter point was brought out in a study by Caramazza and Zurif (1976) who examined comprehension capacities in Broca's, Wernicke's, and conduction aphasics. Patients heard a sentence and then had to choose which of two pictures corresponded to the sentence.

Topic-150: Implications for Understanding Normal Language Processing

A major distinction in the study of normal language processing is between comprehending language and producing it. If this is not merely a conceptual distinction but also biologically based, then we would expect to see some aphasic cases in which comprehension is impaired but production is spared, and vice versa. The initial descriptions of Broca's and Wernicke's aphasias support this distinction. Although there are more subtle differences between these two cases, the comprehension-production distinction appears beyond dispute. Thus, in this instance the studies of aphasia support our distinction. Another intriguing aspect of aphasic language is the sparing of comprehension of axial commands in individuals with Wernicke's aphasia. Although Wernicke's aphasics show many problems in language comprehension, they respond correctly to commands involving midline or axial structures, such as stand up, turn around, take a bow, and look up (Geschwind, 1965). For example, patients perform accurately when asked to stand like a boxer or even take two steps backward, turn around, and sit down again. What is especially striking is that axial commands sometimes involve specific nouns or verbs that these patients ordinarily fail to comprehend.

LATERALIZATION OF LANGUAGE PROCESSES

Topic-151: Split-Brain Research

A consistent finding in the research on aphasia is that language deficits are associated with damage to the left hemisphere of the brain more often than to the right hemisphere. Moreover, we have known for some time, from studies of animals, that communication between the hemispheres may be disrupted by severing the corpus callosum. In the animal studies, one hemisphere could be taught a specific task, and then the other hemisphere could be tested. Typically, little or no learning was found in the other hemisphere, indicating little or no transfer of information between the hemispheres following severing of the corpus callosum. The goal of these studies is to determine what skills are lateralized to one or the other side of the brain.

Topic-152: Lateralization in Normal Brains

Research on the representation of language and nonverbal functions in the intact brain has increased dramatically in the last few decades as techniques for studying the normal brain have been developed and refined. The visual field task used with split-brain patients has also been used frequently with normal individuals. Another commonly used technique, referred to as a dichotic listening task, involves the simultaneous presentation of different stimuli to the two ears. Initially developed by Broadbent (1954) to study attention, the technique was first used by Kimura (1961) to examine ear and hemisphere differences. Kimura (1964) found that recall of verbally presented materials such as digits was superior in the right ear and that recognition of nonverbal stimuli such as melodies was better in the left ear.

Topic-153: Contributions of the Right Hemisphere

So far we have emphasized the talents of the left hemisphere and the ineptness of the right hemisphere. However, the right hemisphere also has some talents in the linguistic realm. Normal individuals use the skills of both hemispheres to comprehend and produce language, so we need to examine some of the ways that the two hemispheres interact during language use. It appears that the right hemisphere is better prepared than the left to appreciate some of the pragmatic aspects of language. Kaplan, Brownell, Jacobs, and Gardner (1990) examined the ability of individuals with right-hemisphere brain damage to interpret conversational remarks. The subjects heard short vignettes that described the performance of one character and the relationship between two characters and then interpreted an utterance from one of the characters. Some of the utterances were literally true, and some were literally false. For instance, in one vignette, Mark was playing golf poorly, and Hal said, you sure are a good golfer, which was literally false. The interpretation of this utterance is in part based on the relationship between the two men. When it was friendly, the comment might be taken as a white lie intended to encourage a friend, but when the relationship was hostile, it could be taken as a sarcastic statement. Kaplan and colleagues found that individuals with right-hemisphere damage were as adept as control subjects in interpreting the literally true sentences but were poorer at identifying the pragmatic intent of literally false utterances. In particular, they had difficulty integrating information about the performance with information about the characters' relationship.

Topic-154: Aphasia in Children and Hemispherectomy Studies

Lenneberg based his hypothesis primarily on studies of children recovering from brain damage. He cites the work of Bassler (1962), who found that if brain damage occurred prior to the onset of speech, speech is often delayed in rate but normal in pattern; children go through the normal stages of language development but proceed more slowly. Bassler also reports that damage to the right hemisphere in the first two years of life produces as much disruption in speech development as damage to the left hemisphere. Bassler (1962) reported that brain damage that is sustained after the onset of speech produces different results. Speech disturbances were considerably more common when the damage occurred in the left hemisphere than in the right hemisphere in a group of children who sustained injuries between 2 and 10 years of age. Injury to the left side resulted in language disturbances nearly twice as often as injury to the right side.

We learn more about the development of lateralization from examining the results of a surgical operation known as a hemispherectomy. This operation is normally used to treat incurable and potentially fatal tumors and involves the removal of either the left or the right cerebral hemisphere. Removal of the right hemisphere in adults leads to little or no language impairment, whereas removal of the left hemisphere leads to significant language problems (Springer & Deutsch, 1998).

Topic-155: Behavioral and Psychological Studies

Some studies have applied behavioral techniques such as dichotic listening to children with normal development. These studies have provided the clearest picture of the development of lateralization to date. Kimura was one of the first to report that children as young as 4 to 6 years could produce adult like right-ear advantages on the dichotic task (Kimura, 1963). Subsequent studies have found that children as young as 2 years of age show the typical right-ear advantage for speech (Springer & Deutsch, 1998). In addition, studies of very young infants have found right-ear advantages for speech as well (Bertoncini, Morais, Bijeljac-Babic, McAdams, Peretz, & Mehler, 1989). The investigators studied infant perception using the high-amplitude sucking procedure. When infants habituated to a particular sound (that is, when their sucking rate decreased), a different sound was substituted. The researchers found that the right ear was better at responding to changes in speech, whereas the left ear was better with changes in musical stimuli.

Topic-156: Development of Lateralization & Lateralization in Other Species

It was once thought that laterality was exclusively human, but we now have several documented cases of the lateralization of species-specific vocalizations. One study (Petersen, Beecher, Zoloth, Moody, & Stebbins, 1978) investigated the perception of vocalizations in Japanese macaque monkeys. The researchers presented the vocalizations to either the left or the right ear. All five macaques showed better performance when the vocalizations were given to the right ear. Only one of five monkeys from other species showed a right-ear advantage for macaque sounds. These results suggest the exciting possibility that human lateralization of speech is part of a larger pattern in which a number of species show lateralization on the left half of the brain for important, species-specific sounds. This view is reinforced by some impressive work on birdsong. Nottebohm (1970) has pointed out that three basic developmental sequences may be observed with young birds. In the first type, the bird develops a normal song even if it is completely isolated at birth and deafened at hatching. Examples of this type are chickens and ring

doves. In a second type, the bird will develop the normal song if isolated but not if deafened. Song sparrows are an example. Finally, in the third variety, either isolation or deafening at an early age produces an abnormal song; chaffinches and white-crowned sparrows fit this pattern. Nottebohm (1970) has found a number of analogs to human speech in this third type of songbird. First, these birds go through a period of “subsong,” similar to human babbling, in which the song is distinct from the adult version in a number of ways. Second, birds in different areas learn different dialects of the same song. Third, the consequences of deafening the bird are different at different ages. If the bird is deafened before it has begun to sing, it develops a highly abnormal song. If, however, the deafening is delayed until after the song has developed, it has no effect on the motor output. Finally, Nottebohm (1970) has found that the left half of the brain is more intimately involved than the right half in the songs of chaffinches. The major structure of the vocal system of the chaffinch is called the syrinx, and each side of it is connected by the hypoglossus nerve to the corresponding side of the brain. (Here, the connection is ipsilateral, with the left hypoglossus connected to the left half of the brain.) Nottebohm found that when the left hypoglossus was severed, the nature of the song was seriously disrupted, with some parts missing and replaced by unstructured bursts of noise. The same operation on the right side leaves the song intact.

EVOLUTION OF LANGUAGE

Topic-157: Evolution of Language

Continuity theories: Continuity theories are built on the idea that language exhibits so much complexity that one cannot imagine it simply appearing from nothing in its final form; therefore it must have evolved from earlier pre-linguistic systems among our primate ancestors. **Discontinuity theories:** Discontinuity theories take the opposite approach—that language, as a unique trait which cannot be compared to anything found among non-humans, must have appeared fairly suddenly during the course of human evolution.

Some theories see language mostly as an innate faculty—largely genetically encoded. Other theories regard language as a mainly cultural system—learned through social interaction.

Topic-158: Communication in Present Day Primates

Noam Chomsky, a prominent proponent of discontinuity theory, argues that a single chance mutation occurred in one individual in the order of 100,000 years ago, installing the language faculty (a component of the mid-brain) in "perfect" or "near-perfect" form. A majority of linguistic scholars as of 2018 hold continuity-based theories, but they vary in how they envision language development. Among those who see language as mostly innate, some—notably Steven Pinker—avoid speculating about specific precursors in nonhuman primates stressing simply that the language faculty must have evolved in the usual gradual way. Others in this intellectual camp—notably Ib Ulbæk—hold that language evolved not from primate communication but from primate cognition, which is significantly more complex.

Topic-159: Teaching Language to Nonhuman Primates

Attempts to teach language (or a language like system of communication) to other primates may be divided into three groups. In the first group, attempts were made to teach speech to chimpanzees (Hayes & Hayes, 1952; Kellogg & Kellogg, 1933). The vocal apparatus of chimpanzees is not suited to produce speech sounds (more on this later); consequently, these studies proved very little. In the Hayes's study, the chimpanzee was able to learn only two words in two years of training. A second group involved programs in which the communication system was not clearly defined in linguistic terms. Premack (1971) used tokens with symbols on them to teach a chimpanzee named Sarah some logical concepts. Although Sarah was able to demonstrate a number of aspects of complex cognition, the results are of uncertain importance to the issue of whether apes can acquire a language because it has not been demonstrated that these symbols have the power or flexibility of human language.

Topic-160: The Continuity Debate

Let us take stock of where we are. It appears that the communication skills of nonhuman primates, studied either in the wild or in the laboratory, fall well short of the full range of human language. In the wild, nonhuman primates display signals that have meaning, but the signals fail to achieve some of the defining characteristics of language. The signals tend to be iconic rather than arbitrary. Moreover, the system of communication is very limited. In human language, we have a duality of patterning that permits us to construct an infinite set of messages from a small set of meaningless

elements. In contrast, the alarm calls of vervet monkeys are global signals that do not consist of elements that can be combined and recombined in different ways. The linguistic skills of chimpanzees and bonobos taught ASL or other language like systems are more impressive at the level of semantics. However, these laboratory animals do not seem to easily grasp the nuances of syntax, something that all normal children attain without specific instruction in their first few years of life. Thus, we are compelled to conclude that human language is qualitatively different than the communication systems of nonhuman primates and, by extension, the common species from which humans, chimpanzees, and bonobos evolved. But this observation creates a dilemma: Darwinian Theory is based on the concept of continuity, the notion that evolutionary changes are quantitative rather than qualitative. As Darwin (1871) commented: “There can be no doubt that the difference between the mind of the lowest man and that of the highest animal is immense....Nevertheless, the difference, great as it is, certainly is one of degree and not of kind” (pp. 127–128). Darwin (1871) also emphasized that the presence of this conclusion is not changed by considering language: ‘Nor ...does the faculty of articulate speech in itself offer any insuperable objection to the belief that man has been developed from some lower form’ (p. 93).

Topic-161: Gesture and Speech as Possible Evolutionary Sequences

One thing seems clear: that spoken language developed quite recently in our evolutionary history. As we saw in chapter 4, when we produce sounds, air is expelled from the lungs and sent through the structures of the vocal tract (see Figure 4.1). We can change the shape of the vocal tract by altering the position of our lips, jaw, tongue, and larynx. The shape and flexibility of our vocal tract are required for the range of sounds found in contemporary languages. A crucial evolutionary development is the enlargement of the pharyngeal area that lies above the larynx and just below the mouth. In newborn humans and adult chimpanzees, the larynx exits directly into the mouth, whereas in adult humans, it exits into the pharynx. This anatomical difference greatly influences the number of sounds that can be produced. For example, newborns and chimpanzees are physically incapable of producing the vowels [a], [i], and [u], sounds that are found in a wide variety of languages. It has often been said that evolution is miserly. The human mouth, throat, and larynx were “designed” for swallowing and breathing, not for speech. In evolutionary terms, speech may be an example of exaptation (Gould, 1980; Gould & Vrba, 1982): It utilizes preexisting physical structures for new functions. In this context, the enlargement of the pharyngeal area has a highly significant consequence. Because in adult humans the passageways are shared among speech, eating, and breathing, the potential for choking on food is much greater than for infants or chimpanzees, which have a smaller pharyngeal area (Lieberman, 1991). Why would such an arrangement have evolved? It is one indication that language is so vital that we have evolved a speech apparatus that increases our capacity to choke in order to use productive speech.

Topic-162: Brain Size and Social Behaviour as Possible Evolutionary Sequences

The finding that brain size increased prior to vocal tract changes helps us pin down the sequence of evolutionary events but also raises an issue. Why did brain size increase? That is, what selective pressures led to this development? As Dunbar (1993, 1998) has pointed out, brain size has costs as well as benefits. It is, for one thing, harder to escape from predators when carrying around such a large brain. Dunbar (1998) asserts that “in the absence of any selection pressure, larger brains will not evolve” (p. 93). Dunbar (1998) suggests that brain size may have increased in relation to group size. As our

evolutionary ancestors began to congregate in larger groups, the mechanisms of group cohesion and regulation began to change. That is, we need a mechanism to hold the larger social group together. With smaller groups, grooming served this essential function, and grooming is a pervasive feature of several nonhuman primates, including chimpanzees. With larger groups, some form of communication may have evolved to play this role. Dunbar claims that language evolved to meet this need by enabling one member of the species to communicate with many members of the group simultaneously.

With regard to our evolutionary ancestors, the social cognition hypothesis is that language evolved as a bonding device. Larger group sizes led to larger brains, including brains more capable of inferring the intentions of others, and larger brains ultimately led to language (Dunbar, 1998). As with all of these views, issues remain to be discussed, but Dunbar's view provides an interesting and plausible explanation of why language evolved. If this view is correct, then social pressure is the driving force behind the evolution of language.

LANGUAGE, CULTURE AND COGNITION

Topic-163: The Whorf Hypothesis

The view that language shapes thought is most often associated with the work of Benjamin Lee Whorf. Whorf received his degree in chemical engineering from the Massachusetts Institute of Technology and worked throughout his life for an insurance company as a fire prevention engineer. He had a number of avocations, however. He had a strong interest in the relationship between science and religion, and ultimately religion led him to language. He was initially self-taught in linguistics but eventually studied American Indian linguistics with the prominent anthropologist Edward Sapir at Yale University. Sapir (1921) had earlier suggested that languages are diverse in the way that they structure reality, but he had not fully developed the thesis that these linguistic differences might facilitate certain modes of thought. This was a position that Whorf developed in a series of articles from 1925 to 1941, many of which are included in Carroll (1956). The notion that language shapes thought patterns is commonly referred to as the Whorf hypothesis, although it is also called the Sapir–Whorf hypothesis, to acknowledge the role of Whorf’s mentor.

Topic-164: Linguistic Determination and Relativity

The Whorf hypothesis consists of two parts: linguistic determinism and linguistic relativity. Linguistic determinism refers to the notion that a language determines certain nonlinguistic cognitive processes. That is, learning a language changes the way a person thinks. Linguistic relativity refers to the claim that the cognitive processes that are determined are different for different languages. Thus, speakers of different languages are said to think in different ways.

We dissect nature along lines laid down by our native languages. The categories and types that we isolate from the world of phenomena we do not find there because they stare every observer in the face; on the contrary, the world is presented in a kaleidoscopic flux of impressions which has to be organized by our minds—and this means largely by the linguistic systems in our minds. We cut nature up, organize it into concepts, and ascribe significances as we do, largely because we are parties to an agreement to organize it in this way—an agreement that holds throughout our speech community and is codified in the patterns of our language. The agreement is, of course, an implicit and unstated one, but its terms are absolutely obligatory; we cannot talk at all except by subscribing to the organization and classification of data which the agreement decrees.

There are several notions here. One is that languages “carve up” reality in different ways. Another is that these language differences are covert or unconscious; that is, we are not consciously aware of the way in which we classify objects. Third, these language differences influence our worldview. These are profound ideas but not ones easily amenable to experimental test. Let us begin by looking, as Whorf did, at linguistic examples from various languages that seem to bear on his thesis.

Topic-165: Some Whorfian Examples: Lexical Examples, Grammatical Examples

Whorf provided a number of examples designed to show that linguistic determinism and relativity were valid concepts. They can be broadly organized into lexical and grammatical examples.

Lexical examples: We may begin by considering the concept of differentiation. Differentiation refers to the number of words in a given domain (for example, colors, birds, fruits, and so on) in a lexicon. A more highly differentiated domain has more words, some of which express finer distinctions, such as subtle shades of color. Whorf argued that languages differ in the domains that are most differentiated. That is, all languages show high degrees of differentiation in some domains and low degrees in others. The implication is that greater degrees of differentiation are related to culturally significant concepts. For example, Whorf noted that in the American Indian language of Hopi, just one word covers everything that flies except birds (for example, the same word for insects, airplanes, aviators, and so on). The Hopi speaker calls all of these disparate objects by the same name without any apparent difficulty. Whorf argued that although this class might seem very broad to us, so would our word snow to an Eskimo:

We have the same word for falling snow, snow on the ground, snow packed hard like ice, slushy snow, wind-driven flying snow—whatever the situation may be. To an Eskimo, this all-inclusive word would be almost unthinkable; he would say that falling snow, slushy snow, and so on, are sensuously and operationally different, different things to contend with; he uses different words for them and for other kinds of snow. (Carroll, 1956, p. 216)

Whorf offered this as evidence of linguistic determinism, of the power of words to influence thought processes. We are careless because of the word empty. Note also that Whorf emphasized “regular” analogy: we come to this (precarious) interpretation of experience based on habitual experience with words. It is something that occurs slowly, over time, seeping into our mental framework. Whorf did not say that we could not avoid this pattern of thought and treat the empty drums with proper respect (obviously he did); but this is a different, more active, pattern of thought.

Topic-166: Grammatical Examples

Although some of Whorf’s lexical examples, such as his comments on Eskimo, have generated a considerable amount of discussion, it appears that he was more interested in the grammatical differences among languages. In English, we come to respect the difference between nouns and verbs as a fundamental distinction. Nouns refer to long-lasting and stable events, such as horse and man, whereas verbs refer to short-lived actions, such as hit and run. Yet, Whorf asked, why then do we classify temporary events such as lightning and spark as nouns? And why are dwell, persist, and continue verbs? In Hopi, lightning is a verb because events of brief duration must be verbs. Whorf also mentioned Nootka, a language used on Vancouver Island, in which all words seem to be treated as verbs. This is just one indication of how grammatical characteristics vary from language to language. Another example of grammatical diversity concerns the extent to which a language uses word order or morphology to signal meaning. In English, the vast majority of sentences use a subject-verb-object (SVO) order, and in most of these the first noun is the agent and the second the patient. This order is adhered to rather rigidly. If the verb is intransitive (one that does not take an object), the remainder of the sequence holds (SV). Similarly, when the first noun is deleted, it is very often replaced by a pronoun; other languages allow deletion of the subject more often. The major exception to the standard order is the question form. The consistency of word order in English makes it a reliable cue for sentence interpretation (MacWhinney, Bates, & Kliegl, 1984). For a speaker of English, languages that violate the SVO order may seem unnatural.

In Wintu, one inflection would be attached to the verb if there were direct visual evidence of this fact, another if it were gossip, and still another if it were a regular event (Lee, 1944/1987). This suggests,

as Brown (1958, p. 254) puts it, that Wintu speakers must have “a continuing grasp of the evidence” of their assertions. Similarly, one cannot directly translate the English sentence ‘I was riding a horse’ into Navaho. Notice that the structure of the sentence is similar to ‘I was pounding a nail’; I is the actor and the nail is the object of the action. In Navaho, the action of riding is shared between the rider and the horse; no person or animal has anything done to him or her or it. The best translation, admittedly awkward, is along the lines of I was animaling-about with the horse (Elgin, 2000). It is more similar to dancing with another person than pounding a nail. Whorf believed that grammatical distinctions such as these exert an effect on not just the way individuals think but also their overall world view. In English, there is a distinction between what Whorf called individual nouns (more commonly called count nouns) and mass nouns. Count nouns refer to bodies with definite outlines (for example, a tree, a stick, a hill), whereas mass nouns refer to objects without clear boundaries (for example, air, water, rain). Linguistically, the distinction is that count nouns take the plural morpheme, whereas mass nouns cannot. Thus, we can speak of trees, sticks, and hills but not airs, waters, and rains. In addition, count nouns take the singular indefinite article a, but mass nouns do not. In contrast, in Hopi, there are no mass nouns. Although we cannot pluralize English mass nouns directly, we can do so by the use of a phrase of the form count noun + of + mass noun. So, even though we cannot say waters or sands, we can say bodies of water or buckets of sand. But this form of expression, according to Whorf, has cognitive consequences because it leads us to think of some objects as being “containers” (form) that hold “contents” (substance or matter). This distinction between form and substance is not a necessary feature of objective reality. For example, even though some objects, such as butter and meat, have clear boundaries, they are treated grammatically as mass nouns (for example, two sticks of butter, not two butters). Thus, Whorf suggested that English speakers think of objects as consisting of form and substance because of this grammatical distinction.

Topic-167: Criticism on Whorf Hypothesis

Any study that attempts to test the hypothesis that differences in language determine differences in thinking must, at the outset, define the three key terms. First, we need to define what we mean by “differences in language.” This has been done in two ways. One way is to compare a language that linguistically marks a particular conceptual distinction with a language that does not. Thus, the presence or absence of the explicit linguistic marking is the language difference of interest. Although most studies have approached the problem in this way, another possibility is to compare two languages that mark the same distinction in different ways. This comparison focuses not on whether a language marks a concept but rather how it does so. As we have seen, English marks number through the use of the plural morpheme. One comparison would be another language that does not mark number; another would be a language that marks number in a different way. Second, we need to define “differences in thinking” in a satisfactory manner. It is obviously difficult to measure a person’s world view. But it should be kept in mind that Whorf was especially interested in those aspects of thinking that indicated a habitual mode of thought. Lucy (1992b) defines habitual thought as “routine ways of attending to objects and events, categorizing them, remembering them, and perhaps even reflecting upon them” (p. 7). The mode is contrasted with specialized thought, which is composed of cognitive routines or structures that are restricted either to certain subgroups within a culture (such as technical specialists) or to certain domains (such as kinship or illness).

Topic-168: Sapir-Whorf Hypothesis: Limitations and Possibilities

Some sources indicate that the recent work in linguistics and cognitive science has supported the Sapir-Whorf hypothesis. *Language in Mind: Advances in the Study of Language and Thought* (2003) is a collection of papers that gives three ideas on language and thought concepts.

Some studies conclude that the ideas put forth by Daniel Chandler, Gerooge Grace, and Lawrence Campbell indicate the Sapir-Whorf hypothesis being significant, but not being applicable to all situations. A *Linguistic Mystic* (Notes from a Linguistic Mystic) explains why Sapir-Whorf is not conclusively proved and disproved. The Mystic writes everybody views the world differently and uses their language accordingly. The language of the one from a teaching career is different from the language of the other who is from a photographic printing business. Similarly, there might be individual cognitive (and linguistic) quirks with every person. The result is that there is no neutral sample of a given language. Culture is another important factor to be considered under discussion. People's views of the world might depend upon their cultural norms, beliefs, and perceptions. Similarly, people from the same given native language or dialect might share the same cultural background, so it is difficult to find out whether the given effect or consequence has resulted from linguistic or cultural perspectives or the perspectives from both. Such things might get the researcher into a "which came first, the chicken or the egg" type debate. Last but not the least would be the issue of the experiment itself. When one tries to study how people use language, without biasing them, he uses language to explain the study and conduct the experiments. Then he might need a translator to pass on instructions, which may bias the participant right from the get-go. The Mystic adds that if there is some degree of linguistic relativity, it will likely be universal, and thus, the researcher will be influenced by it, too. In such a case, looking at the nature of these effects or consequences, a researcher studying this effect or consequence in another person, might be like an inmate studying the behavior of fellow inmates.

THEORIES ON CULTURE AND LANGUAGE

Topic-169: Theories on Language & Culture.

Theory 1: Speech is Essential for Thought

Following are the theories given by different researchers on language and culture. (We must learn how to speak aloud, otherwise we cannot develop thinking.)

Proponents:

a) Thought is a kind of behavior, speech, which originates from speech production (verbal or non-verbal)

b) Thought develops as a kind of speech

- By speaking aloud, you start to speak subvocally or make internal articulations. (Thought is defined as subvocal speech or behavior.)

Inadequacies of the theory:

1) Children having no speech production can comprehend and think.

E.g. Hearing persons who are born mute.

- If a person can comprehend the meaning of speech, that person must have the ability to think.

2) Speech comprehension, which implies thought, develops from speech production in normal children

- Speech comprehension precedes speech production. In other words, speech comprehension is the basis for speech production.

E.g. Children who could only produce single-word utterances could understand syntactic structures composed of more than one word. This shows that the children's level of speech comprehension is well in advanced of their level of speech production, which is an only one-word production.

3) Simultaneously speaking aloud while thinking about something different.- Thinking of something other than what we are saying (wandering mind).

4) Telling a lie

- Saying one thing while thinking something quite different. This means that when we tell lies, two distinct processes (speaking aloud and speaking subvocally/to yourselves) with different content occur at the same time.

5) Meaning and thought occur without behaviour

- People do not lose meaning of words when any parts of their body are lost/removed.

E.g. Paralysed person can still think clearly.

- This shows that thought was not dependent on body movements or movements of the speech organs.

6) Interpreting between languages can be done

- Consider the work of simultaneous interpreters. They have to understand (a process done in their minds) the messages told to them in one language and then transform the language to another language by

speaking it aloud.

- This shows that a system of abstract thought to mediate between languages has made simultaneous interpretation possible.

All of these 6 objections to the theory show that speech production is not necessary for thought.

Topic-170: Theory 2: Language is Essential for Thought

Language is essential for thought (We must learn language, how to produce or understand speech, otherwise we cannot develop thinking).

Proponents:

- a) The language system, with its rule or vocabulary, is necessary for thought.
- b) Thought was derived from speech production.
- c) Thought is supposed to be language-specific and not universal.

Inadequacies of the theory:

1) Deaf persons without language can think.

E.g. Deaf children, when at play and when participating in activities around the home, behave as intelligently and rationally with respect to the environment as do hearing children.

- If one holds that language is the basis for thought, then these deaf children do not think and that they were merely robots.

2) Multilinguals are whole persons.

- According to this theory, if multilinguals have more than one thought process (one for each language), such persons would not be able to think coherently or would have separate thought intelligencies/personalities. Persons knowing 3 languages would have formed 3 systems of thought, for example.

- This is not true because it is proved through a case that a multilingual and a monolingual child living in the same environment have no difference in terms of beliefs, values, personality and their perceptions of the world and nature.

3) Intelligent animal behaviour occurs without language.

- Thought must have some basis other than language. The following examples can prove this statement.

E.g.:

- a) Monkeys (without language) can develop some simple counting and arithmetic abilities.
 - b) Insects (also without language) can think of strategies for gathering food, defending themselves, and finding mates.
- Since animals can think without language, there is no reason to believe that humans cannot do the same.

Topic-171: Theory 3: Language Determines or Shapes Our Perception of Nature

Language determines or shapes our perception of nature (The learning of language will determine or influence the way we perceive the physical world, visually, auditorily, etc.)

Proponent

- One's knowledge of vocabulary or syntax influences one's perception and understanding of nature.

Inadequacies of the theory

1) Perception, interest, and need determine vocabulary

- It is our interest and need that determine our coinage of vocabulary and its use.

E.g. Children, from all over the world, are enchanted by dinosaurs. They perceive the types of dinosaurs. Through perception, they develop their interesting dinosaurs and later they feel the need to seek the names of these objects.

2) Color and snow vocabulary

- Rather than language determining perception, it is perception that determines language.

- Color words

E.g. Speakers of a language with limited repertoire of color terms appeared no different from speakers with broader repertoire of color terms in terms of distinguishing colors of rainbow.

- Snow words

E.g. Hawaiians have only one, the English word 'snow' but the Inuits have single words for snow-on-the-ground, hard-snow-on-the-ground, block-of-snow and others. As for English-speaking skiers in cold countries, they name snow through its physical condition by creating phrases namely 'powder snow', 'wet snow', etc.

- It is because of the importance of snow in their lives that they have created more words for snow than have Hawaiians.

- It is this language device of creating phrases which every language has that makes up for any vocabulary deficiency.

3) Hopi 'Time' and Chinese 'Counterfactuals'

- Hopi people and time

- People are not different because of their language, but because of their experiences. Deep down, we are all the same; it couldn't be otherwise.

E.g. Hopi people use periods relating to the harvest, the moon, the sun and other significant events. We do much the same in English (".....when it gets dark", "when the weather gets warm").

- The Chinese language and 'counterfactuals'

- Chinese were not as able as English speakers to think hypothetically about what is not true because of certain grammatical features of the Chinese language.

- This happened due to faulty translations but once proper translations were made, there was no basis for claiming a difference in thinking.

4) Lack of vocabulary does not indicate lack of concept

- We describe a thing, which does not have a single word for it, with a phrase carrying a similar concept.

E.g., we have a name for the underside of our hand that is called 'palm' but we have no word for the topside. Instead we use the phrase 'back of the hand.'

- This shows that lack of vocabulary item is not indicative of a lack of a concept.

5) Knowledge overrides literal word meanings

- We can believe something quite different from what the language literally specifies and that the continual use of a language form may not change an underlying thought. In other words, one thing is said but another is understood.

(similar to lying, but in this case, people know it is not true)

E.g., the word 'sunset'. We always hear and use this word that it leads us to believe that the sun sets on its own. The truth is, it is the earth that moves, not the sun.

6) Multilingual's view of nature

- If it is said that different languages have distinctive and important effects on the way we view nature, then the multilingual must similarly have distinctive and important ways of viewing nature. But such is not the case. Multilingual is a whole person who perceives nature as other humans do.

Topic-172: Theory 4: Language Determines or Shapes our cultural world view

Language determines or shapes our cultural world view (The learning of language will determine or influence the way we understand our culture and the world).

Proponents

a) Even if language is somewhat distinct from thought, nevertheless, knowing a language will itself condition and influence one's cultural, social beliefs or views of the world.

b) Language does provide a view of culture and society and an outlook on the world.

Inadequacies of the theory

1) Same language yet different world views

- According to the theory, speakers of the same language must have same world views. This is not true.

E.g. A monolingual English-speaking family of members with various religions (Mother is a Buddhist, father is a Christian, son is a Moslem), view the world differently according to the beliefs of their religions.

2) Different languages yet similar world views

E.g., Communist Doctrine is shared by speakers of many languages namely Koreans, Chinese, Vietnamese, and Cubans.

3) Same language but world view changes over time

- Changes in world view occur without changes in language.

- Changes in world view can occur due to causes other than the language system namely the spread of new ideas, economic domination, wars, etc.

4) One language can describe many different world views

E.g., the Bible of the ancient Hebrew people.

- If this Bible were to be translated into many languages with perfect translations, the meaning of the Bible in every language may differ because every word in any languages carry different primary or secondary meanings, implications, presuppositions, feelings, attitudes, and politeness.
- This will cause many different world views out of a Bible of Hebrew language.

5) Multilingual's world view

- The theory predicts that a multilingual will have as many distinct world views as language systems. This is, again, not true because as been mentioned before, multilingual is a whole person who perceives nature as other humans do.

Topic-173: Erroneous Beliefs Underlying the Four Theories

Erroneous belief 1: their analysis of language is adequate: The most serious deficiency in the theorizing of Whorf, Sapir, Korzybski, Skinner, von Humboldt, and others concerns the assumption that the directly observable words or the structure of a sentence represent all of the semantic or thought elements of that sentence. These theorists drew conclusions largely based on what linguists today would consider a superficial surface structure analysis. Whorf (Carroll, 1956), for example, states that 'Our Indian languages show that with a suitable grammar we may have intelligent sentences that cannot be broken into subjects and predicates' (p. 242), and that 'the Hopi language contains no reference to "time" either explicit or implicit' (p. 58). Such statements (disavowed by subsequent linguists) are made essentially because it is assumed that surface structure directly signifies all of the meaning of a sentence. These linguists thus had a tendency to focus on the differences between languages. The idea that grammatical differences made thought differences was not long in coming. It seems that most of the linguists simply played with the idea, because they offered little or no evidence for their assertions. They had other concerns. Except for Whorf, they did not bother to seek out hard evidence to support their contentions. It was only when Chomsky came along and postulated underlying structures that it was realized that languages had a lot of similarities. It is not surprising, therefore, that only after Chomsky did the search for language, universals begin in earnest, although the linguist Joseph Greenberg was an earlier initiator.

Erroneous belief 2: The meaning of words is linguistic in origin except for the minor case of onomatopoeia (speech sounds imitating environmental sounds): the relationship between a word and its meaning is conventional. That is, **there is no necessary relationship between the sound of a word and its meaning.** There is no inherent reason for a dog to be called 'dog' and not 'pen.' Thus, when one hears a word for the first time, e.g. the obscure English word 'tantivy,' its meaning is not understood, unless it is composed of known morphemes. The meaning that is to be associated with a particular sound sequence must be learned. It is not possible to know from the sound sequence alone (especially if not in a phrase or sentence) that the meaning of 'tantivy' is related to horses and indicates 'a gallop or rapid movement.' Meaning for words is acquired in four main ways: (1) a sound form is associated with an object, situation, or event in the world, e.g. the sound 'dog' with the object 'dog'; (2) a sound form is

associated with an idea or experience in the mind, e.g. 'pain' with the feeling of 'pain'; (3) an inference may be made in a linguistic context, an idea may be suggested, e.g. in reading a paragraph one word may not be known but because everything else is understood, its meaning may be guessed at by inference; and (4) an analysis of known component morphemes may suggest a meaning for the sound form, e.g. the meaning of 'unprimitive' can be gained through knowledge of the morphemes 'un' and 'primitive.' In considering these four ways of acquiring word meaning, we may note that the first two involve non-linguistic sources. In (1) the experiencing of objects, situations, and events in the world provides a basis for meaning, and in (2) experiences in the mind itself provide the basis. The mistake of Whorf and others is to assume that the mere hearing of the sound form of a word (an unknown one) itself provides some sort of meaning. They do not allow for a prior mental experience because that would imply that thought precedes language. A language sound form itself, however, does not provide meaning.

Erroneous belief 3: There are primitive languages and primitive human intelligence: One often hears observers of other cultures say that such and such a people are not logical or rational or that they are somehow deficient in intellect. Such supposed deficiencies are frequently attributed to their having a primitive language. Yet modern linguistic research has never found a single language that could be called 'primitive.' Thus it is that Chomsky (1967a) can with some assurance assert that all languages are of similar complexity, with each having similar basic forms and operations. And, while other linguists may disagree with Chomsky on what the basic form and operations might be, they do not disagree that all languages are constructed and operate with essentially the same principles. It appears to us that Sapir, Whorf, and the others assumed that there was such a thing as a primitive people and that those people had a primitive grammar. Since they supposed that the primitive grammar of these people reflected their thought, they concluded that their thought was primitive too. How else could Whorf have concluded that the Hopi people had no concept of time, even with the faulty linguistic data that he gathered from those people? Such notions about primitive peoples and primitive languages were virtually taken for granted in the first half of the twentieth century when Sapir and Whorf were doing their theorizing, when the world still had undiscovered peoples and lands. Those days seemed long gone. Then, too, there is a case of non-standard speakers of such dialects as Black English, whom many naive people regard as irrational. Labov (1970), however, clearly demonstrated that logic does underlie the utterances of those speakers. Once one learns the premises that a people hold, their behaviour and statements that were previously thought to be strange or illogical immediately become rational.

Topic-174: The Best Theory: Thought is Independent of Language

The thought system in the mind of the child develops over time as input stimuli of the world, such as visual, auditory, and tactile stimuli representing objects, events, and situations in the environment, are experienced by the child. Until thought is sufficiently developed (ideas of objects, relations of objects, states and actions of objects), words uttered in the presence of the child are not meaningfully processed. When that happens and when language input is experienced in coordination with objects, events, and situations, then language can begin to be learned. Over a period of time, the language system, with its vocabulary and grammatical rules, is formed. Part of the language system is actually part of the thought system, for the meaning and semantics of the language system are those ideas that are part of the content of thought. There is not one idea for 'dog' in language and another in thought. Such a view would be unparsimonious in the extreme. Rather, the thought and language systems are joined through meaning and ideas.

LEXICAL INFLUENCES ON COGNITION

Topic-175: Testing the Whorf Hypothesis

Any study that attempts to test the hypothesis that differences in language determine differences in thinking must, at the outset, define the three key terms. First, we need to define what we mean by ‘differences in language.’ This has been done in two ways. One way is to compare a language that linguistically marks a particular conceptual distinction with a language that does not. Thus, the presence or absence of the explicit linguistic marking is the language difference of interest. Although most studies have approached the problem in this way, another possibility is to compare two languages that mark the same distinction in different ways. This comparison focuses not on whether a language marks a concept but rather how it does so. As we have seen, English marks number through the use of the plural morpheme. One comparison would be another language that does not mark number; another would be a language that marks number in a different way.

Second, we need to define “differences in thinking” in a satisfactory manner. It is obviously difficult to measure a person’s world view. But it should be kept in mind that Whorf was especially interested in those aspects of thinking that indicated a habitual mode of thought. Lucy (1992b) defines habitual thought as “routine ways of attending to objects and events, categorizing them, remembering them, and perhaps even reflecting upon them” (p. 7). The mode is contrasted with specialized thought, which is composed of cognitive routines or structures that are restricted either to certain subgroups within a culture (such as technical specialists) or to certain domains (such as kinship or illness).

Topic-176: Color Terms

Codability: A concept that has figured in much of the research on color cognition is codability. Brown (1958; see also Lenneberg, 1953) defined codability as the length of a verbal expression. As we saw in our discussion of differentiation, some languages have single words to refer to a particular object or event, whereas others do not. If one’s language does not have a specific word for the occasion, the speaker can still make the reference but will need to do so by some combination of words. Relative to the case in which a single word serves the purpose, the phrase is, in Brown’s terms, less codable. Brown (1958) suggested a relationship between the frequency of usage of a verbal expression, its length (codability), and the ease with which it may be used. The relationship between frequency and length is captured in what is called Zipf’s law. Some time ago, Zipf (1935) examined Chinese, Latin, and English and found that the length of a word is negatively correlated with its frequency of usage. That is, the more frequently a word is used in a language, the shorter the word (measured either in phonemes or syllables). English contains many examples of Zipf’s law. Whenever mass-produced technological innovations are introduced in society, their initial, cumbersome names become shortened for easy reference (for example, video camera-videocassette recorder becomes camcorder). It may be that the differences in the differentiation of domains that Whorf observed are a special instance of Zipf’s law. For instance, it may be that in cultures in which an object is referred to extremely often, it is referred to by a single, brief name; when it is moderately frequent, by a longer name; and when it is infrequent, by a phrase. The relationship between codability and ease of expression has been studied in several experiments. In an early study, Brown and Lenneberg (1954) examined the responses of college students to 24 different

colors. The colors were identified beforehand by a set of judges who were asked to look at a series of color chips and determine which the best instance of the color in question was. The judges produced a list of 8 central colors, with 16 other colors included for comparison. The 24 colors were shown to the students, one at a time, and they were asked to name the colors, with their reaction time to naming the colors being measured. Brown and Lenneberg found that colors that evoked long names (that is, those less codable) were named with hesitation, with disagreement from one person to another, and with inconsistency from one time to another.

Topic-177: Cross-Linguistic Studies

These results suggest that the presence of a brief verbal expression in a language influences certain cognitive processes. However, to evaluate the notion of linguistic determinism, we need to study the effects of color terms in different languages. Berlin and Kay (1969) have investigated color terms in various languages. They found that although the number of color terms in a language varied quite a bit from language to language, there was an underlying order. They found that every language has a small number of basic color terms. These are terms that consist of only one morpheme (for example, blue versus blue-green), are not contained within another color (for example, crimson is contained within the category of red), and are not restricted to a small number of objects (for example, blond is restricted mainly to hair color). Furthermore, each language draws its basic color terms from the following list of 11 names: white, black, red, yellow, green, blue, brown, purple, pink, orange, and gray. In addition, Berlin and Kay found that these 11 terms formed a hierarchy.

On balance, results suggest that under some circumstances, the manner in which we perceive and remember colors is related to the linguistic terms we use to refer to them. Thus, the color domain appears to provide support for the weak version of linguistic relativity.

Topic-178: Number Terms

Another set of studies is relevant to how the lexicon may influence thought processes. These studies have examined how morphological differences in number names between Asian languages (Chinese, Korean, and Japanese) and English may influence children's conceptualization of numbers and, ultimately, their mathematics achievement. The linguistic distinction here is not whether the different languages name numbers but how they do so. In English, the system of naming numbers is relatively complex. The names for numbers 11 and 12, for instance, are unrelated to the names for 1 and 2. The names for 13 through 19 consist of the unit name before the decade name (for example, seventeen). Furthermore, the names for numbers between 20 and 99 consist of the decade name followed by the unit name (for example, thirty-three). In contrast, Asian languages such as Chinese are more regular. The names for numbers between 11 and 99 consist of the decade name followed by the unit name. For instance, the Chinese word for 18 is ten eight and the word for 35 is three ten five. For numbers less than 10 and greater than 99, English and Chinese naming systems are more similar (Hurford, 1987). The greater regularity of Asian languages suggests that children might have an easier time acquiring number names than their English-speaking counterparts. Some evidence indicates that this is so. Miura (1987) studied first-grade children from the United States and Japan. The children were shown how to use a set of base-ten blocks to represent numbers. The set consisted of white unit blocks and purple tens blocks equivalent to ten unit blocks stuck together. On the first trial, the children were asked, in their native language, to read a number on a card and then to show that number in the blocks. After doing this for 5

minutes, they were given a second trial. They were reminded of the equivalence of ten unit blocks and one tens block and were then asked to show each number in another way. Miura distinguished three approaches to the task. A canonical approach was one that placed no more than nine unit blocks in the one's position, such as using four tens blocks and two unit blocks for 42. A noncanonical approach was one that used some combination of tens blocks and more than nine unit blocks, such as three tens blocks and 12 units blocks for 42. Finally, a one-to-one collection used only unit blocks, such as 42 unit blocks. The results indicated that Japanese children were more than twice as likely as U.S. children to use canonical approaches on the first trial. The U.S. children tended to use one-to-one collections on the first trial. When prodded to generate a second approach, the U.S. children developed canonical approaches on the second trial. Miura also found that Japanese children used more noncanonical approaches than U.S. children. Miura has found similar results using Korean and Chinese first graders (Miura, Kim, Chang, & Okamoto, 1988). Furthermore, as predicted from the analysis of the naming systems, the range of numbers between 11 and 99 shows the greatest differences. Chinese preschoolers are no better than American preschoolers at counting between 1 and 10 or beyond 99 (although few can do the latter). But Chinese children are better at counting between 11 and 99 than their English-speaking counterparts (Miller, Smith, Zhu, & Zhang, 1995). Additional work suggests that Chinese speakers pronounce numbers more quickly than English speakers (Hoosain, 1986; Hoosain & Salili, 1987) and that there is a correlation between speed of number pronunciation and mathematical performance (Ellis & Hennelly, 1980). Similarly, Miura and Okamoto (1989) found that Chinese children understood place value better and had a higher level of mathematics performance than English-speaking children. This finding would suggest that a strategy of teaching place value would improve mathematics performance, and there is some evidence to support this line of thought. Fuson, Smith, and Lo Cicero (1997) found that explicit teaching of the base-ten concept improved computational performance in low-achieving Latino first graders in the United States. The studies converge on the conclusion that the way that languages represent numbers influences mathematical thinking. The differences in mathematics achievement between Asian children and American children have been well documented (Stevenson, Lee, & Stigler, 1986), and there are surely many contributing factors to this difference, including parental emphasis, pedagogical techniques, and broader cultural influences. Nonetheless, the language one learns plays a role in mathematics education. Miura stresses that the way one thinks about numbers is fundamentally different in Chinese versus English. That is, it is not simply that Asian children do better on mathematics tasks; they appear to approach the tasks differently as well.

Topic-179: Object Terms

Recent research in how infants learn names pertaining to objects is also relevant here. You may recall the studies in Chapter 10 that discussed the relationship between object permanence and language acquisition. The conclusion drawn by some researchers (e.g., Gopnik, 2001) was that conceptual categories related to object names are constructed at the time when we learn a language, not before. If so, then we might expect to see different kinds of early object terms in children acquiring different languages. Gopnik and Choi (1990) examined the linguistic and cognitive development of Korean-speaking children. Compared to English, Korean uses fewer nouns and permits noun ellipsis, particularly when it is contextually obvious what is being referred to (Clancy, 1985). Gopnik and Choi found that compared to English children, Korean children were delayed in categorization tasks and the naming explosion. A subsequent study (Gopnik & Choi, 1995; Gopnik, Choi, & Baumberger, 1996) found that Korean-speaking children were superior to English-speaking children in means-ends abilities and success/failure

words. In contrast, the English speakers were superior in categorization and the naming spurt. This pattern of data appears to be related to the observation that Korean-speaking mothers used more verbs and fewer nouns than English-speaking mothers (Gopnik, Choi, & Baumberger, 1996). Thus, it appears that the prevalence of nouns and verbs in speech given to children (as well as the way they are used; see Gopnik, 2001) may influence the timing of certain cognitive achievements.

Topic-180: Spatial Terms

Bowerman and Choi (2001) discuss the question posed at the beginning of the chapter—what came first, cognition or language—and arrive at an interactionist view. Children’s early word meanings are neither simply labels for existing concepts (the cognitive view) nor constructed entirely because language requires it (the Whorf hypothesis). Rather, they result from the interaction of existing cognitive development and the semantic categories of the input language. Bowerman and Choi (2001) review evidence on the acquisition of spatial terms and conclude that there are substantial similarities across languages. The order of acquisition of spatial terms is relatively consistent. Some terms (such as behind or in front) tend to be underextended and others (such as open) tend to be overextended, and this pattern is also consistent across languages. Also, some spatial words are generalized rather rapidly. All of these results suggest that children may have a rich knowledge of space prior to learning the specific spatial terms their native language encodes. At the same time, Bowerman and Choi (2001) document considerable cross linguistic variation in spatial terms. For example, English and Korean differ substantially. English makes a fundamental distinction between putting on and putting in. Korean uses a term (kkita) to mark a property that we are not familiar with in English: whether two objects with complementary shapes fit together into an interlocking, tight-fitting relationship; this applies to such disparate cases as putting a VHS tape into a case, putting a piece in a puzzle, stacking LEGOs, and putting a ring on a finger. Notice that in English, the first two of these would be considered putting in and the other two putting on. In effect, Korean makes the distinction between putting things into tight containers or loose containers, such as putting an apple in a bowl. Choi and Bowerman (1991; Bowerman, 1996) compared early acquisition of spatial terms in Korean and English and found significant differences. Although both groups of children began using spatial terms around 14 to 16 months of age, they used spatial terms in different ways. English-speaking children distinguished between putting things into containers and putting them onto surfaces, but paid no attention to whether the object fit the container tightly or loosely. The Korean learners, in contrast, distinguished between tight and loose containment. Another cross-linguistic study used the comprehension time paradigm pioneered by Golinkoff and Hirsh-Pasek. In this study, children between 18 and 23 months of age already seem to understand kkita. Because most of the children are not yet producing these words, it appears that sensitivity to language-specific grammatical properties is underway even before language production begins (Choi, McDonough, Bowerman, & Mandler, 1999). It is clear from these studies that acquiring the semantics of Korean influences Korean infants’ conceptualization of the world. But the nature of the influence is not entirely clear. Does the presence of the linguistic terms create cognitive categories, as supposed by the strong version of the Whorf hypothesis? Or do such categories exist prior to language experience and are then enhanced or diminished by language experience? Hespos and Spelke (2004) suggest the latter view is correct. They examined how 5-month-old infants in an English-speaking environment responded to the “tight” versus “loose” distinction.

GRAMMATICAL INFLUENCES ON COGNITION

Topic-181: Grammatical Influences on Cognition: Studies of Subjunctive

H. Bloom (1981) has conducted some interesting but controversial studies on the differences between how Chinese and English speakers reason. He was particularly interested in counterfactual reasoning, which is the ability to reason about an event that is contrary to fact. For instance, imagine a situation in which several people are waiting for John; he is late, and the group, as a consequence, is late to the movies. The English language has the subjunctive mood, shown in sentence (2), which enables us to discuss various states of affairs that we know to be false: (2) If John had come earlier, they would have arrived at the movies on time (adapted from Bloom, 1981, p. 19). We also need to consider problems on the cognitive side. As noted earlier, Whorf was principally interested in habitual modes of thought. Lucy (1992b) suggests that counterfactual reasoning is more specialized than habitual because it is probably more accessible to those with higher levels of education. It thus remains to be seen whether Whorfian effects can be observed when more habitual forms of thought are assessed.

Topic-182: The Development of Subjunctive and Complex-Syntactic

For many foreign-language (FL) learners of Spanish, one of the most unique grammatical constructs of the Spanish language is the subjunctive. The subjunctive is not highly productive in English, and so students have almost no L1 models with which to formulate hypotheses about its use in Spanish. This challenge is augmented by the subjunctive's linguistic complexities, denoting abstract concepts (e.g., marking irrealis events and states) and having a syntactic distribution that is largely limited subordinate clauses. The present review critically examines almost thirty years of research aimed at understanding the processes through which learners pass in the development of subjunctive abilities and the factors that interact with those processes.

While the primary purpose of this chapter is to provide an understanding of subjunctive acquisition in FL contexts, it contextualizes this foray with an analysis of important studies conducted on the acquisition of the subjunctive in first-language (L1) and bilingual contexts. Subsequently, the author examines the factors that affect subjunctive development in FL contexts as well as the variables influencing the development of abilities for processing complex syntax. Finally, the chapter outlines experimental design issues that limit the generalization of the conclusions that these L1, bilingual, and FL studies have offered.

Topic-183: Grammatical Marking of Form

A study by Carroll and Casagrande (1958) compared Navaho and English. They observed that in Navaho, the form of the verb for handling an object varies with the form or shape of the object. The verb varies if the object is a long flexible object (such as a piece of string) versus a long rigid object (such as a stick) or a flat flexible object (such as a cloth). On the basis of this grammatical distinction, Carroll and Casagrande proposed that Navaho-speaking children would learn to discriminate the forms of objects at an earlier age than their English-speaking peers. Carroll and Casagrande (1958) used an object triads test in which the child had to pick which of two objects, of three presented, went best together. For example, a child might be presented with a yellow stick and a piece of blue rope of comparable size. The child would

then be shown a yellow rope and asked with which of the first two it went best. Thus, it is possible to determine whether the children were focusing more on color or form. Carroll and Casagrande compared children who spoke Navaho with children who spoke English but came from the same reservation and lived in similar circumstances. These results seem generally to support Whorf's view that the grammatical distinctions in a language may influence or determine certain cognitive processes. But the observations from the suburban children suggest that even if grammatical categories determine qualities of thought, they are not the only determinants. Other attributes of the child's environment may serve the same function.

Topic-184: Grammatical Marking of Objects and Substances

Languages also differ in their grammatical distinction of objects and substances. As we have already seen, in English objects such as candles and chairs are referred to as count nouns. Count nouns have distinct singular and plural forms (for example, candle, candles). Moreover, count nouns can be counted (one candle, two candles, and so on). In contrast, nouns such as air, water, and mud are referred to as mass nouns. They cannot take the plural morpheme and cannot be directly counted. We can count even mass nouns by using expressions such as two buckets of mud or three gallons of water. In contrast, in Japanese, all inanimate nouns are treated like English mass nouns (Gentner & Boroditsky, 2001). These nouns cannot take the plural morpheme and can only be counted in the indirect manner of English mass nouns. What, then, does a prelinguistic child see when looking at, say, a rock and a pile of mud? Does the child see these things as fundamentally different as compared to, say, a rock and a stick? If the Whorf hypothesis is correct, children would not notice these similarities and differences before acquiring the linguistic distinction (between mass and count nouns) that draws one's attention to them. Moreover, the hypothesis would suggest that children acquiring English and Japanese would see this aspect of the world very differently.

Topic-185: Grammatical Marking of Gender

English marks grammatical gender only in singular personal pronouns (for example, he, she, it). In contrast, other languages have much more extensive gender systems. Spanish nouns that refer to males end in -o (as in hermano or brother, and gato or male cat) and words that refer to females end in -a (hermana, gata). In addition, many Spanish nouns are marked for gender even if there is no obvious semantic basis for it. For example, key, grapes, and table are feminine in Spanish, and plane, telephone, and bucket are masculine. It is clear that the gender systems of English and Spanish are quite different, and thus it is natural to investigate what effect these systems might have on their learners.

Topic-186: Final Observations

Both Sapir and Whorf agreed that it is our culture that determines our language, which in turn determines the way that we categorize our thoughts about the world and our experiences in it.

For more than fifty years, researchers have tried to design studies that will support or refute this hypothesis. Support for the strong version has been weak because it is virtually impossible to test one's world view without using language. Support for the weaker version has been minimal.

Problems with the hypothesis begin when one tries to discern exactly what the hypothesis is stating. Penn notes that the hypothesis is stated "more and less strongly in different places in Sapir's and

Whorf's writings" (1972:13). At some points, Sapir and Whorf appear to support the strong version of the hypothesis and at others they only support the weak version. Alford (1980) also notes that neither Sapir nor Whorf actually named any of their ideas about language and cognition the Sapir-Whorf Hypothesis. This name only appeared after their deaths. This has led to a wide interpretation of what researchers consider to be the one and only hypothesis.

Another problem with the hypothesis is that it requires a measurement of human thought. Measuring thought and one's world view is nearly impossible without the confounding influence of language, another of the variables being studied. Researchers settle for the study of behaviour as a direct link to thought.

If one is to believe the strong version of linguistic determinism, one also has to agree that thought is not possible without language. What about the pre-linguistic thought of babies? How can babies acquire language without thought? Also, where did language come from? In the linguistic determinist's view, language would have to be derived from a source outside the human realm because thought is impossible without language and before language there would have been no thought.

NEUROLINGUISTICS AND DISORDERS

Topic-187: Neurolinguistics and Disorders: Disorder of Syntax

Syntactic deficits are common in language disorders and have always been at the focus of research on language disorders. The investigation whether or not syntactic deficits occur in a given acquired or developmental language disorder, which syntactic structures or processes are eventually affected and how to capture such deficits in an explanatory theoretical account has dominated the linguistic research on language disorders since its very first beginnings to the present. Interest on syntactic deficits in language disorders first focused on Broca's aphasia – an acquired language disorder caused by strokes affecting left frontal brain regions. The core symptom of Broca's aphasia is a so-called agrammatic spontaneous speech production which is characterized by omissions of free functional elements and a severe reduction of the length and syntactic complexity of utterances. This leads to a preponderance of very short utterances that seem to consist of a simple, linearly organized string of open-class words. As the term 'agrammatic' already indicates, the spontaneous speech of language impaired speakers suffering from this disorder gives the impression of lacking syntactic structure. Hence, agrammatic Broca's aphasia has been considered the ideal candidate for investigating syntactic deficits and has been at the focus of linguistic investigations concerned with syntactic disorders ever since the advent of modern generative syntactic theory (Chomsky 1957). According to Chomsky's generative approach, the human language faculty is due to a specialized, domain-specific language organ situated in the brain that is part of our biological endowment and therefore genetically specified (e.g. Chomsky 1980, 2002). If this conception holds true, the language organ in the human brain should be affected by brain lesions involving this organ. If the impaired brain tissue subserves this language organ, the impairment of this organ due to a brain lesion should lead to impairments of the language faculty. Agrammatic Broca's aphasia seemed to exemplify this theoretically predicted case of an impaired language faculty due to brain lesions. Consequently, from the late 1960's to the late 1980's, the linguistic investigation of language disorders associated with syntactic deficits was predominantly concerned with the language deficits observed in Broca's aphasia, the focus of this research being on the issue whether or not Agrammatism is due to a deficit affecting syntactic competence and how to capture this purported syntactic-competence deficit in a syntax-theoretical framework.

Topic-188: Aphasia to Neurolinguistics

Aphasia is an acquired language disorder subsequent to brain damage in the left hemisphere. It is characterized by diminished abilities to produce and understand both spoken and written language compared with the speaker's presumed ability pre-cerebral damage. The type and severity of the aphasia depends not only on the location and extent of the cerebral damage but also the effect the lesion has on connecting areas of the brain. Type and severity of aphasia is diagnosed in comparison with assumed normal adult language. Language changes associated with normal aging are not classed as aphasia. The diagnosis and assessment of aphasia in children, which is unusual, takes account of age norms.

The most common cause of aphasia is a cerebral vascular accident (CVA) commonly referred to as a stroke, but brain damage following traumatic head injury such as road accidents or gunshot wounds can also cause aphasia. Aphasia following such traumatic events is non-progressive in contrast to aphasia

arising from brain tumor, some types of infection, or language disturbances in progressive conditions such as Alzheimer's disease, where the language disturbance increases as the disease progresses.

The diagnosis of primary progressive aphasia (as opposed to non-progressive aphasia, the main focus of this article) is based on the following inclusion and exclusion criteria by M. Marsel Mesulam, in 2001. Inclusion criteria are as follows: difficulty with language that interferes with activities of daily living and aphasia is the most prominent symptom. Exclusion criteria are as follows: other non-degenerative disease or medical disorder, psychiatric diagnosis, episodic memory, visual memory, and visuo-perceptual impairment, and, finally, initial behavioral disturbance.

Topic-189: Reading and Writing Disorders

Reading disorders occur when a person has trouble with any part of the reading process. Reading and language-based learning disabilities are commonly called dyslexia. These disorders are present from a young age and usually result from specific differences in the way the brain processes language.

There are many different symptoms and types of reading disorders, and not everyone with a reading disorder has every symptom. People with reading disorders may have problems recognizing words that they already know and may also be poor spellers. Other symptoms may include the following:

- Trouble with handwriting
- Difficulty reading quickly
- Problems reading with correct expression
- Problems understanding the written word

Reading disorders are not a type of intellectual and development disorder, and they are not a sign of lower intelligence or unwillingness to learn.

People with reading disorders may have other learning disabilities, too, including problems with writing or numbers.

Topic-190: Phonological and Surface Dyslexia

Surface dyslexia is a type of dyslexia, or reading disorder. According to Marshall & Newcombe's (1973) and McCarthy & Warrington's study (1990), patients with this kind of disorder cannot recognize a word as a whole due to the damage of the left parietal or temporal lobe. Individuals with surface dyslexia are unable to recognize a word as a whole word and retrieve its pronunciation from memory. Rather, individuals with surface dyslexia rely on pronunciation rules. Thus, patients with this particular type of reading disorder read non-words fluently, like "yatchet", but struggle with words that defy pronunciation rules (i.e. exception words). For example, a patient with surface dyslexia can correctly read regular words like "mint" but will err when presented a word that disobeys typical pronunciation rules, like "pint". Often, semantic knowledge is preserved in individuals with surface dyslexia.

Phonological disorder: Individuals with that form of dyslexia typically have difficulty sounding out unfamiliar words and do poorly on tests of non-word reading. Students can't break down individual sounds of language (phonemic awareness) and match them with written symbols.

This makes it difficult to sound out or "decode" words. Most kids with reading issues have some degree of phonological dyslexia. It's also sometimes referred to as dysphonetic dyslexia.

Topic-191: Dyslexia

Dyslexia is a brain-based type of learning disability that specifically impairs a person's ability to read. Individuals with dyslexia typically read at levels significantly lower than expected despite having normal intelligence. Although the disorder varies from person to person, common characteristics among people with dyslexia are: difficulty with phonological processing (the manipulation of sounds), spelling, and/or rapid visual-verbal responding. Dyslexia can be inherited in some families, and recent studies have identified a number of genes that may predispose an individual to developing dyslexia. Examples of specific types of reading disorders include:

- **Word decoding:** People who have difficulty sounding out written words; matching the letters to sounds to be able to read a word.
- **Lack of fluency:** People who lack fluency have difficulty reading quickly, accurately, and with proper expression (if reading aloud).
- **Poor reading comprehension:** People with poor reading comprehension have trouble understanding what they read.

Topic-192: Deep Dyslexia

Deep dyslexia is usually classified as an acquired reading disorder, as opposed to developmental dyslexia, in previously literate adults as a consequence of a brain injury. However, recently, developmental deep dyslexia has also been reported in children with Williams's syndrome.

Deep dyslexia is considered to be "central dyslexia" as compared to "peripheral dyslexia." Peripheral dyslexics have difficulty matching the visual characteristics of letters that comprise a word to a stored memory of this word from prior encounters. Central dyslexics are unable to properly match the visual word to the word's meaning. They may also be incapable of speaking, or phonating, the sequence of written letters that they see into the word these letters represent. Deep dyslexia differs from other forms of central dyslexia (phonological dyslexia and surface dyslexia) in that deep dyslexics have many more symptoms and these symptoms are generally more severe. According to the "continuum" hypothesis, deep dyslexia is a more severe form of phonological dyslexia.

Deep dyslexia is mainly characterized by the occurrence of semantic reading errors or semantic paralexias (transposition of letters or words) when reading aloud (e.g. the written word "view" is read aloud as "scene", the word "bird" is read as "canary." These semantic errors are the major distinguishing feature of deep dyslexia in comparison to other central dyslexias. There are many other symptoms of deep dyslexia, including visual errors (e.g. the written word "thing" is read aloud as "think", the word "skate" is read as "scale") and derivational errors (e.g. the written word "alcohol" is read aloud as "alcoholic", the word "governor" is read as "government"), and poor reading of function words. Additionally, deep dyslexics have more difficulty reading abstract than concrete and highly imaginable words, more difficulty reading adjectives, adverbs, and verbs than nouns, a complete inability to read non-words, and often impairments on tasks of verbal working memory.

FIRST LANGUAGE ACQUISITION

Topic-193: Theories of First Language Acquisition

Various theories have arose since language studies came to fore, and the ability to acquire language has interested various parties since the dawn of man. From the dunes of Egypt Psammeticus, the Pharaoh during the 7th century BC, believed language was inborn and that children isolated from birth from any linguistic influence would develop the language they had been born with. Fast forward to the 15th century, King James V of Scotland performed a similar experiment; the children were reported to have spoken good Hebrew. Akbar, a 16th century Mogul emperor of India, desired to learn whether language was innate or acquired through exposure to the speech of adults. He believed that language was learned by people listening to each other and therefore a child could not develop language alone. So he ordered a house built for two infants and stationed a mute nurse to care for them. The children did not acquire speech, which seemed to prove Akbar's hypothesis that language is acquired and does not simply emerge spontaneously in the absence of exposure to speech.

Henceforth, modern linguists have been trying hard to crack the codes which govern the acquisition and learning of a language. Theories ranging from Jean Piaget's Cognitive Theory(1929), Skinner's Behaviorist Theory (1957), to Chomsky's The Innateness Hypothesis, and Lambert's Critical Period Hypothesis(1967) for first language acquisition, and finally Krashen's 5 hypothesis of second language learning have paved a way for an insight, a way to unravel the way the mind works in acquiring and learning a language -which happen to be distinct from one another- and here, we will be looking at the theories that have been the workhorse of language acquisition and learning.

Topic-194: Challenges of Approaches to First Language Acquisition

Skinner's theories attracted a number of critics, not the least among them Noam Chomsky (1959), who penned a highly critical review of Verbal Behavior. Some years later, however, KemieUi MacCorquodale (1970) published a reply to Chomsky's review in which he eloquently defended Skinner's points of view. And so the controversy raged on.

Some psychologists proposed modified theoretical positions. One of these positions was mediation theory, in which meaning was accounted for by the claim that the linguistic stimulus (a word or sentence) elicits a "mediating" response that is self-stimulating. Charles Osgood (1953, 1957) called this self-stimulation a "representational mediation process," a process that is really covert and invisible, acting within the learner. It is interesting that mediation theory thus attempted to account for abstraction by a notion drat reeked of "mentalism"—a cardinal sin for dyed-in-the-wool behaviorists! In fact, in some ways mediation theory was really a rational/cognitive theory masquerading as behavioral. Mediation theories still left many questions about language unanswered. The abstract nature of language and the relationship between meaning and utterance were unresolved. AH sentences have deep structures—deep level of underlying meaning that is only manifested over by surface structures. These deep structures are intricately interwoven in a person's total cognitive and affective experience. Such depths of language were scarcely plumbed by mediation theory.

Topic-195: Issues in First Language Acquisition

Competence and performance: For centuries scientists and philosophers have drawn basic distinction between competence and performance. Competence refers to one's underlying knowledge of a system, event, or fact. It is the non-observable ability to do something, to perform something. Performance is the overtly observable and concrete manifestation or realization of competence. It is the actual doing of something: walking, singing, dancing, speaking. In technological societies we have used the competence-performance distinction in all walks of life. In our schools, for example, we have assumed that children possess certain competence in given areas and that this competence can be measured and assessed by means of the observation of elicited samples of performance called "tests" and "examinations." In reference to language, competence is one's underlying knowledge of the system of a language—its rules of grammar, its vocabulary, all the pieces of a language and how those pieces fit together. Performance is actual production (speaking, writing) or the comprehension (listening, reading) of linguistic events. Chomsky (1965) likened competence to an "idealized" speaker-hearer who does not display such performance variables as memory limitations, distractions, shifts of attention and interest, errors, and hesitation phenomena such as repeats, false starts, pauses, omissions, and additions. Chomsky's point was that a theory of language had to be a theory of competence lest the linguist try in vain to categorize an infinite number of performance variables that are not reflective of the underlying linguistic ability of the speaker-hearer.

Comprehension and production: Not to be confused with the competence-performance distinction, comprehension, and production can be aspects of both performance and competence. One of the myths that has crept into some foreign language teaching materials is that comprehension (listening, reading) can be equated with competence, while production (speaking, writing) is performance, it is important to recognize that this is not the case; production is of course more directly observable, but comprehension is as much performance as production is. In child language, most observational and research evidence points to the general superiority of comprehension over production; children seem to understand "more" than they actually produce. For instance, a child may understand a sentence with an embedded relative in it (e.g., "The ball that's in the sandbox is red") but not be able to produce one. W. R. Miller (1963, p. 863) gave us a good example of this phenomenon in phonological development: "Recently a three-year-old child told me her name was Litha. I answered 'Litha?' 'No, Litha.' 'Oh, Lisa.' 'Yes, Litha.'" The child clearly perceived the contrast between English s and th, even though she could not produce the contrast herself. How are we to explain this difference, this apparent "lag" between comprehension and production? We know that even adults understand more vocabulary than they ever use in speech, and also perceive more syntactic variation than they actually produce. Could it be that the same competence accounts for both modes of performance? Or can we speak of comprehension competence as something that is identified as separate from production competence? Because comprehension for the most part runs ahead of production, is it more completely indicative of our overall competence? Is production indicative of a smaller portion of competence? Surely not. It is therefore necessary to make a distinction between production competence and comprehension competence. A theory of language must include some accounting of the separation of the two types of competence. In fact, linguistic competence no doubt has several modes or levels, at least as many as four, since speaking, listening, reading, and writing are all separate modes of performance.

Topic-196: Nature or Nurture Universal

Nativists contend that a child is born with an innate knowledge of or predisposition toward language, and that this innate property (the LAD or UG) is universal in all human beings. The innateness hypothesis was a possible resolution of the contradiction between the behavioral notion that language is a set of habits that can be acquired by a process of conditioning and the fact that such conditioning is much too slow and inefficient a process to account for the acquisition of a phenomenon as complex as language. But the innateness hypothesis presented a number of problems itself. One of the difficulties has already been discussed in this chapter: the LAD proposition simply postpones facing the central issue of the nature of the human being's capacity for language acquisition. Having thus "explained" language acquisition, one must now scientifically explain the genetic transmission of linguistic ability—which we cannot yet do with certainty. And, of course, scholars taking an emergentist perspective continue to challenge the notion that what is innate is grammatical or linguistic at all. On the other hand, while the LAD remains a tentative hypothesis, I think we can take heart in slowly mounting genetic (scientific) evidence of the transmission of certain abilities, and assume that among those abilities we will one day find hard evidence of "language genes." We must not put all our eggs in the innateness basket. Environmental factors cannot by any means be ignored, as connectionists and emergentists have shown. For years linguists, psychologists, and educators have been embroiled in the "nature-nurture" controversy: What are those behaviors that "nature" provides innately, in some sort of predetermined biological timetable, and what are those behaviors that are, by environmental exposure—by "nurture," by teaching—learned and internalized? We do observe that language acquisition is universal, that every child acquires language. But how are the efficiency and success of that learning determined by the environment the child is in? Or by the child's individual construction of linguistic reality in interaction with others? The waters of the innateness hypothesis are considerably muddled by such questions. An interesting line of research on innateness was pursued by Derek Bickerton (1981), who found evidence, across a number of languages, of common patterns of linguistic and cognitive development. He proposed that human beings are "bio-programmed" to proceed from stage to stage. Like flowering plants, people are innately programmed to "release" certain properties of language at certain developmental ages. Just as we cannot make a geranium bloom before its "time," human beings will "bloom" in predetermined, preprogrammed steps.

Topic-197: Language and Thought Imitation. Practice and Frequency. Input. Discourse

For years researchers have probed the relationship between language and cognition. The behavioral view that cognition is too mentalistic to be studied by the scientific method is diametrically opposed to such positions as that of Piaget (1972) who claimed that cognitive development is at the very center of the human organism and that language is dependent upon and springs from cognitive development. Others emphasized the influence of language on cognitive development. Jerome Bruner (Bruner, Olver, & Greenfield, 1966), for example, singled out sources of language-influenced intellectual development: words shaping concepts, dialogues between parent and child or teacher and child serving to orient and educate, and other sources. Vygotsky (1962, 1978) also differed from Piaget in claiming that social interaction, through language, is a prerequisite to cognitive development. Thought and language were seen as two distinct cognitive operations that grow together (Schinke-Llano, 1993).

Topic-198: First Language Acquisition Insights Applied to Language Teaching

There is a strong evidence for an innate linguistic capacity that assists in the acquisition of linguistic structures. However, by broadening the conception of language to include semantics and pragmatics, the child's environmental and social interactions become increasingly important. The capacity to learn language is therefore multifarious and due to a seemingly inextricable combination of biological and environmental factors. It is with this in mind that the essay will turn to consider whether similar language learning capacities exist within second language learners, noting from the outset some difficulties that arise in making cross-learner comparisons.

AGE AND ACQUISITION

Topic-199: Children Vs. Adults in Second-Language Learning

The comparison of first and second language acquisition can easily be oversimplified. At the very least, one needs to approach the comparison by first considering the differences between children and adults. It is, in one sense, illogical to compare the first language acquisition of a child with the second language acquisition of an adult (Foster-Cohen, 2001; Scovel, 1999; Schachter, 1988; Cook, 1973). This involves trying to draw analogies not only between first and second language learning situations but also between children and adults. It is much more logical to compare first and second language learning in children or to compare second language learning in children and adults. Nevertheless, child first language acquisition and adult second language acquisition are common and important categories of acquisition to compare. It is reasonable, therefore, to view the latter type of comparison within a matrix of possible comparisons. In general, however, an adult is considered to be one who has reached the age of puberty. There have been few recorded instances of an adult acquiring a first language. In one widely publicized instance, Curtiss (1977) wrote about Genie, a 13-year-old girl who had been socially isolated and abused all her life until she was discovered, and who was then faced with the task of acquiring a first language. Accounts of "wolf children" and instances of severe disability fall into this category.

Topic-200: Neurobiological Considerations Hemispheric Lateralization

Hemispheric Lateralization: Some scholars have singled out the lateralization of the brain as the key to answering such a question. There is evidence in neurological research that as the human brain matures, certain functions are assigned, or "lateralized" to the left hemisphere of the brain, and certain other functions to the right hemisphere. Intellectual, logical, and analytic functions appear to be largely located in the left hemisphere, while the right hemisphere controls functions related to emotional and social needs. Language functions appear to be controlled mainly in the left hemisphere, although there is a good deal of conflicting evidence. For example, patients who have had left hemispherectomies have been capable of comprehending and producing an amazing amount of language. Generally, a stroke or accident victim who suffers a lesion in the left hemisphere will manifest some language impairment, which is less often the case with right hemisphere lesions. However, before drawing any conclusions here, some caution is in order. Millar and Whitaker's (1983, p. 110) conclusion of over 20 years ago still stands: "enough data have accumulated to challenge the simple view that the left hemisphere is the language hemisphere and the right hemisphere does something else." While questions about precisely how language is lateralized in the brain are interesting indeed, a more crucial question for second language researchers has centered on when lateralization takes place, and whether or not that lateralization process affects language acquisition. Uric Lcnneberg (1967) and others suggested that lateralization is a slow process that begins around the age of 2 and is completed around puberty. During this time the child is presumably neurologically assigning functions little by little to one side of the brain or the other; included in these functions, of course, is language.

Topic-201: Anthropological Evidence

Anthropological Evidence: Some adults have been known to acquire an authentic accent in a second language after the age of puberty, but such individuals are few and far between. Anthropologist Jane Hill (1970) provided an intriguing response to Scovel's (1969) study by citing anthropological research on non-Western societies that yielded evidence that adults can, in the normal course of their lives, acquire second languages perfectly. One unique instance of second language acquisition in adulthood was reported by Sorenson (1967), who studied the Tukano culture of South America. At least two dozen languages were spoken among these communities, and each tribal group, identified by the language it speaks, is an exogamous unit; that is, people must marry outside their group, and hence almost always marry someone who speaks another language. Sorenson reported that during adolescence, individuals actively and almost suddenly began to speak two or three other languages to which they had been exposed at some point. Even more recently, Mover (2004) has reminded us of a multitude of cognitive, social, psychological, and strategic variables affecting the ultimate attainment of proficiency in a second language.

Cognitive considerations

Human cognition develops rapidly throughout the first 16 years of life and less rapidly thereafter. Some cognitive changes are critical; others are more gradual and difficult to detect. Jean Piaget (1972; 1955; Piaget & Inhelder, 1969) outlined the course of intellectual development in a child through various stages: * Sensorimotor stage (birth to 2) * Preoperational stage (ages 2 to 7) * Operational stage (ages 7 to 16) • Concrete operational stage (ages 7 to 11) * Formal operational stage (ages 11 to 16). A critical stage for a consideration of the effects of age on second language acquisition appears to occur, in Piaget's outline, at puberty (age 11 in his model).

It is here that a person becomes capable of abstraction, of formal thinking which transcends concrete experience and direct perception. Cognitively, then, an argument can be made for a critical period of language acquisition by connecting language acquisition and the concrete/formal stage transition. However, as reasonable as such a contention might sound, even here some caution is warranted. Singleton and Ryan offer a number of objections to connecting Piagetian stages of development with critical period arguments, not the least of which was the "vagueness" and lack of empirical data in Piaget's theory. Ausubel (1964) hinted at the relevance of such a connection when he noted that adults learning a second language could profit from certain grammatical explanations and deductive thinking that obviously would be pointless for a child. Whether adults do in fact profit from such explanations depends, of course, on the suitability and efficiency of the explanation, the teacher, the context, and other pedagogical variables.

Topic-202: Interference Between First and Second Languages

The only way a learner can start to communicate in a second language is the time a learner begins to assume word-for-word translation equivalence or it is thought that every L1 word has one translation in L2 by the learners (Blum-Kulka & Levenston, as cited in Bhela, 1999, p. 30). When learners of second language want to write or speak in the target language, they tend to rely on their first language structures. If the structures are different, then a lot of errors occur in L1. Thus this indicates an

interference of first language on second language. Interference is the errors that can be traced back to the first language, while the learners use the second language.

Topic-203: Issues in First Language Acquisition Revisited

Issue one: Position meeting the criteria for a scientific theory: Chomsky's innateness position seems to meet a convincing criterion for a scientific theory. It covers aspects of some other theories, possesses the properties of scientific theory and has claims for contents. Chomsky believes that both body and mind exist, and the elements needed for theory-making are body, behavior, and mind. His position shares aspects from other positions; for example, epiphenomenalism considers mind as the side effect of an action.

Issue two: Child's logical thinking: Piaget supports the child's logical thinking which is attributable to his third stage of development called "concrete operational period" (occurring roughly around the age of 7 to 11 years). Piaget labels stage three as "preparation for an achievement of concrete operations."

Issue three: The origins of language: First, we refer to Chomsky's position and then to that of Vygotsky. Chomsky is nativist, and he argues that the ability to acquire language is innate and that children are programmed to learn language. Some form of pre-programming can explain the speed with which children learn the complex skill of using language and the similarity of language acquisition across cultures.

Topic-204: Order of Acquisition

The order of acquisition is a concept in language acquisition describing the specific order in which all language learners acquire the grammatical features of their first language. This concept is based on the observation that all children acquire their first language in a fixed, universal order, regardless of the specific grammatical structure of the language they learn. Linguistic research has largely confirmed that this phenomenon is true for first-language learners; order of acquisition for second-language learners is much less consistent. It is not clear why the order differs for second-language learners, though current research suggests this variability may stem from first-language interference or general cognitive interference from nonlinguistic mental faculties.

There is research to suggest that most SLA learners begin their learning process with a silent period in which the learners begin to process pieces of the language they hear. This is considered a period of "language shock", in which they ignore some of the incomprehensible input of the new language. However, research has shown that many "silent learners" are engaging in private speech, sometimes called "self-talk." While appearing silent, they are rehearsing important survival phrases (lexical chunks). These memorized phrases are soon used in various situations, either by choice or necessity. Fewer learners have no silent period and pass directly into patterned speech. This speech is used to accomplish basic communication, often showing few departures from detached words strung together, which in time leads to more fluid phrases. Grammar of the target language is also simplified and the learners begin to construct an understanding of the second language, often attempting "sentences" that mix words or phrases from both their first and second languages.

CHILDREN VS. ADULTS IN SECOND-LANGUAGE LEARNING

Topic-205: Children are Better: A Common Belief Psychological Category

Speaking a second language is an important skill for all people, both young and old. It has long been believed that children are better able to learn a second language. In actuality, it is not that children learn language better than adults, but that adults and children learn language differently. By understanding these differences and making adjustments to the learning process, all people can acquire a second language, no matter their age.

Processing differences: There are distinct differences in the way an adult brain processes a foreign language when compared to the brain of a child. Dr. Paul Thompson of UCLA used MRI imaging and animation technology to view what parts of the brain adults and children use when learning a second language. What was found is that children use a part of their brain called the “deep motor area.” The “deep motor area” of the brain is responsible for processes that are not consciously thought about, like brushing your teeth or getting dressed. For children, processing a new language is second nature. Adults process language in a more active part of the brain, meaning that they think more consciously about language rather than it being intuitive.

Proficiency differences: One reason that it seems that children acquire a second language quicker than adults is because of the different standards of proficiency between adults and children. Children have a smaller vocabulary and it is easy to learn enough of a second language to communicate their needs. Adults have a much larger vocabulary and think and communicate in more complex ways than children. This means it takes them longer to acquire the ability to communicate effectively in a second language. Although it seems that children learn language quicker than adults, in actuality adults and adolescents have the edge.

Pronunciation: Another reason that the myth persists that children learn second languages easier than adults is because of the child’s ability to adapt the proper pronunciation of a language. It is true that the younger a child begins to learn a second language, the better their pronunciation. Adults have a more difficult time adapting the pronunciation of a foreign language, and so sound less competent than a child who has the ability to speak a second language with the proper accent.

Aging and learning ability: Another common misconception is that as people age, their ability to learn a new skill diminishes. In actuality, people do not lose their ability to learn as they age. The only challenges an older learner of a second language faces is the weakening of vision and hearing. The loss of hearing in particular can affect a person’s ability to learn a language in the traditional classroom setting. A healthy, older adult is perfectly capable of learning a second language.

Learning methods: Because children and adults learn differently and use different parts of their brains to process language, the way they are taught a second language should also differ. Exposing children to a second language at home as well as at school is essential to their learning. Singing songs, reading books, and repetition of foreign words are all useful tools in helping a child learn a new language.

Older learners, especially those with hearing and vision difficulties, may have difficulties learning in a traditional classroom setting. Working with a group that focuses more on understanding the language rather than perfecting pronunciation, and integrates new concepts into the adults preexisting cognitive structures will help the older learner succeed.

Motivation: A number of factors that affect second-language learning operate only in certain types of situations. The question of motivation for learning a second language, for instance, is not likely to arise in a natural type of setting such as with a young child. A 1- or 2-year-old needs no motivation to learn a second language; given language input, the young child will automatically learn – with learning even occurring in negative circumstances. An older child of 4 or 5 years, however, may need motivation in order to learn a second language since by that age the child may be aware of whether a language is positively or negatively regarded by others, or the child may prefer other activities.

Attitude: A negative attitude towards the target language or its speakers, or the other members of the class, may also affect one's determination and persistence to be involved in the classroom and its activities. This same negative attitude could impair memory functioning and detract from focusing on the target language. In the same way, any of a host of personality and sociocultural variables could have deleterious effects (Brown, 1987).

Topic-206: Social Category

Three types of social structures which can affect the acquisition of second languages are: sociolinguistic setting, specific social factors, and situational factors. Sociolinguistic setting refers to the role of the second language in society. Specific social factors that can affect second language acquisition include age, gender, social class, and ethnic identity. Situational factors are those which vary between each social interaction.

Language attitudes: Language attitudes in the learner, the peer group, the school, the neighborhood, and society at large can have an enormous effect on the second language learning process, both positive and negative.

Learner attitudes: Learners manifest different attitudes towards 1) the target language, 2) the target language speakers, 3) the target language culture, 4) the social value of learning the L2, 5) particular use of the target language, and 6) themselves as members of their own culture. In general, positive attitudes towards the L2, its speakers, and its culture can enhance learning, which can in turn be influenced by this success; negative attitudes can impede learning. However, if learners have a strong reason for learning, negative attitude can have a positive effect. Learners may have conflicting attitudes. Learners might want to learn the L2 as a way of assimilating into the majority culture, but at the same time, they may wish to keep their L1 as a means of maintaining their L1 identity.

Age: Younger speakers are subject to peer group pressure, (and they use the nonstandard form used by their peer group); the middle-aged group is less subject to peer group pressure, and they are more influenced by mainstream societal values. For older people, the social pressures lessen and social networks again become narrow. Generally, learners who learn an L2 after puberty (or possibly earlier) are unlikely to acquire a native-speaker accent while those who begin after 15 years are less likely to develop full grammatical ability – Chambers and Trudgill (1980).

Gender: Two apparently contradictory principles noticed by sociolinguists are:

1. In stable sociolinguistic stratification, men use a higher frequency of nonstandard forms than women.
2. In the majority of linguistic changes, women use a higher frequency of the incoming forms than men.

Women may be better language learners. They are more sensitive to the new linguistic forms and are more ready to incorporate them into their speech. Thus they will be more likely to rid themselves of any inter language forms that deviate from target-language norms.

In sum, currently there are no clear-cut explanations yet as to why females outperform males in L2 learning. The reasons that females hold a more positive attitude seems to have been widely agreed. Male and female culture differences are also possible explanations. Females are more cooperative and more delicate in dealing with relationship while males emphasize more maintaining their hierarchical relationship. Thus female 'culture' fits L2 learning, as it is more readily to deal with the threat to their identity posed by L2 learning. There is also some evidence suggesting that females' better listening comprehension ability puts them at an advantage in L2 learning.

Topic-207: Basic Psychological Factors Affecting Second-Language Learning

Explication: The nature of explication is the process whereby the rules and structures of a second language are explained to a learner. This explanation is given in the first language of the learner. The learner is then expected to understand, learn, and apply the rule in the second language. Why a language cannot be learned completely by explication while parts of a second language can be learned by explication, it is impossible for it to be learned entirely by explication. This is because not all of the rules of any one language have been discovered and written down. Even for a language such as English, the most researched of all languages; one still finds linguistic journals discussing the concepts involved in such commonplace features of English as tense and the article.

Induction: Learning rules by self-discovery is the essence of the process of induction. The child who is exposed to second-language speech and remembers what he or she has heard will be able to analyze and discover the generalization or rule that underlies that speech. Actually, not only must the learner devise the rule based on the speech that has been heard, but he or she must also figure out how those rules are to be applied in other cases.

Memory: Vocabulary learning and rote memory is crucial to learning. It is inconceivable that a person with severe memory impairment could ever learn his or her native language, much less a second language. The learning of the simplest word requires memory. A person learning the word 'dog', for example, must retain a connection between the hearing of 'dog' and the experience of seeing, touching, or smelling a dog. Such a connection between the sound and the object is arbitrary. There is no logical relationship between the sound 'dog' and its meaning.

Topic-208: Summary of Three Important Psychological Factors Affecting Second-Language Learning

Let us now summarize the effects on second-language learning of the various psychological variables. Three basic psychological categories are: Intellectual, which is subdivided into Inductive and Explicative; Memory; and Motor Skills. Along the left margin of the table, persons are divided into three age groups: Children under 7, Children 7 to 12, and Adults over 12.

Induction: We can see that insofar as Induction is concerned, this ability remains at a relatively high level with age, except with certain individuals in old age. Such ability allows us to make new discoveries.

Topic-209: Social Situations Affecting Second-Language Learning

There are many social situations in which a second language is learned. Basically, we can cover the most important of them according to three categories, the natural, the classroom, and community context. The natural situation in which a second language is learned is one that is similar to that in which the first language is learned. It can involve social situations such as those involving family, play, or the workplace. The classroom situation involves the social situation of the school classroom. Each of these types of social situations has its own advantages and disadvantages. The community context allows students to have access to a natural situation outside of the class and thereby supplement their classroom learning.

Topic-210: Classroom Situations: Is There A Critical Age for Second-Language Learning?

Adults can learn a second language; it is reasonable to ask the same question about the acquisition of a second language. Is there any barrier to the learning of a second language and, if so, at what age does this barrier become operational? As far as adult second-language learning is concerned, we have the common observation that a very great number of adults do, in fact, learn the syntax of other languages perfectly. There are those who speak second languages so well that, on the basis of the grammar alone (not the pronunciation, which we shall deal with shortly), they would be judged native speakers. There is no demonstrated critical age for learning syntax. There are, however, studies which demonstrate a differential effect for the age at which acquisition of syntax began. Patkowski (1980) had native speakers of English rate the syntax of transcripts of spontaneous speech from immigrants to the USA who had entered before or after the age of 15. Transcripts were used to remove any possible influence of accent on the raters.

LANGUAGE, LEARNING, AND TEACHING

Topic-211: Second Language Acquisition: Learner Characteristics, Linguistic Factors, Learning Processes

Three learner characteristics have consistently been found to be consequential for language learning: motivation, anxiety, and beliefs about language learning.

Topic-212: Age and Acquisition Instructional Variables

The age of second language (L2) acquisition is a factor that has raised a lot of interest and controversy. This phenomenon is called the “critical period” or “time sensitive” According to Lenneberg, the critical period for language acquisition begins around the age of two. Research published prior to the mid-1990s claimed that people learning a second language after puberty still retain a foreign accent, while those who acquired it before puberty did not. Scovel suggests that if second language learning begins after the age of 12 years, learners can never “pass themselves off as native speakers phonologically”.

In early childhood, becoming bilingual is often an unconscious event, as natural as learning to walk or ride a bicycle. But why? According scientific surveys, language aspects such as **pronunciation** and **intonation** can be acquired easier during childhood, due to **neuromuscular mechanisms** which are only active until to the age of 12. Long agrees, but goes further by conditioning the acquisition of a native competence in morphology and syntax to exposure to the second before the age of 15. These assertions are contradicted by empirical studies on older beginners who have reached very high in second language levels of competence.

Other factors that we should take into consideration are children’s **flexibility**, **spontaneity** and **tolerance** to new experiences. Kids are more willing to communicate with people than adults, they are curious and they are not afraid of making mistakes. They handle difficulties (such as missing vocabulary) very easily by using creative methods to communicate, such as non-verbal means of communication and use of onomatopoeic words. Also the idea of a foreign civilization is not formed in their minds yet. Only at the age of 8 does it become clear to them that there are ethnic and cultural differences. Last but not least, aspects such as time, greater **learning and memory capacity** are in any case advantages in early language learning. On the other hand there are surveys which point out the risk of **semi-legalism** and advice parents to organize language planning carefully.

Topic-213: Rejoicing in Our Defeats: Language Learning and Teaching

Second language learning

Second language learning is a conscious process where the learning of another language other than the First language (L1) takes place. Often confused with bilingualism and multilingualism, the process has to take place after the first language(s) has already been acquired. Having said that, Second language learning could also refer to the third, fourth, or fifth (so on and so forth) language the learner is currently learning. People who adopt the memory strategy depend on their **memorizing ability**. They find ways to remember better to aid in entering information into long-term memory, by creating a word-

meaning map in their brain (mental linkages), and then being able to retrieve that information. Adopting this strategy will allow the learning and retrieval via sounds (e.g., rhyming), images (e.g., a mental picture of the word itself or the meaning of the word), a combination of sounds and images (e.g., the keyword method), body movement (e.g., total physical response), mechanical means (e.g., flashcards), or location (e.g., on a page or blackboard). Things they do: do a lot of exercises on English grammar. Create a word bank from your reading materials or TV shows and memorize the meaning of the words and try to use them.

Cognitive strategy

People who adopt the cognitive strategy tend to **analyze and reason**. They form internal mental codes and revise them to receive and produce the message in the target language. Adopting this strategy will enable you to internalize the language in direct ways such as through reasoning, analysis, note-taking, summarizing, synthesizing, outlining, and practicing in naturalistic settings, and practicing structures and sounds formally.

Things they do: People learning Korean watch Korean dramas and try to replicate how the characters pronounce Korean words. Watch Korean dramas and try to replicate how the characters use certain words in a sentence. Write emails or letters in SL. Read SL reading materials such as magazines and newspapers.

Comprehension strategy

People who adopt the comprehension strategy find themselves **guessing unknown words when listening and reading**. They also try to replace words they do not know with longer phrases or other words that they know when speaking and writing to overcome gaps in knowledge.

Things they do: Try to guess the meaning of words they don't know. Try to understand the meaning through looking at the word in context. Guess the meaning of some words by reading the whole passage. Try to look for cues or nonverbal signs when in conversation.

Metacognitive strategy

People who adopt the metacognitive strategy **plan, arrange, focus, evaluate on their own learning process**. They identify and monitor their own learning style preferences and needs, such as gathering and organizing L2 materials, arranging a study space and a schedule for L2 revision and learning, monitoring mistakes made in L2, evaluating task success, and evaluating the success of any type of learning strategy.

Things they do: Observe how the SL teacher speaks in the SL. Observe how they themselves speak in the SL. Practice speaking in SL in front of the mirror. Crosscheck with Google to find out if their pronunciation is correct, and correct it. Doing crossword puzzles and play word games like scrabble. Take note of how other people communicate in SL, especially natives.

Language teaching

Teaching cannot be defined apart from learning. Teaching is guiding and facilitating learning, enabling the learner to learn, setting the conditions for learning. Your understanding of how the learner learns will determine your philosophy of education, your teaching style, your approach, methods, and classroom techniques. If, like B. F. Skinner, you look at learning as a process of operant conditioning

through a carefully paced program of reinforcement, you will teach accordingly. If you view second language learning as a deductive rather than an inductive process, you will probably choose to present copious rules and paradigms to your students rather than let them "discover" those rules inductively. An extended definition—or theory—of teaching will spell out governing principles for choosing certain methods and techniques. A theory of teaching, in harmony with your integrated understanding of the learner and of the subject matter to be learned, will point the way to successful procedures on a given day for given learners under the various constraints of the particular context of learning. In other words, your theory of teaching is your theory of learning "stood on its head."

Topic-214: Schools of Thought in Second Language Acquisition

Structural Linguistics and Behavioral Psychology: In the 1940s and 1950s, the structural, or descriptive, school of linguistics, with its advocates—Leonard Bloomfield, Edward Sapir, Charles Hockett, Charles Fries, and others—prided itself in a rigorous application of scientific observations of human languages. Only "publicly observable responses" could be subject to investigation. The linguist's task, according to the Structuralist, was to describe human languages and to identify the structural characteristics of those languages. An important axiom of structural linguistics was that languages can differ from each other without limit, and that no preconceptions could apply across languages. Freeman Twaddeli (1935, p. 57) stated this principle in perhaps its most extreme terms: whatever our attitude toward mind, spirit, soul, etc., as realities, we must agree that the scientist proceeds as though there were no such things, as though all his information were acquired through processes of his physiological nervous system. Insofar as he occupies himself with psychical, nonmaterial forces, the scientist is not a scientist. The scientific method is quite simply the convention that mind does not exist. Twaddeli was underscoring the mandate for the structural linguist to examine only overtly observable data, and to ignore the "mind" insofar as the latter represented a rationalistic approach that gave credence to unobservable guesses, hunches and intuition. Such attitudes prevailed in B. E Skinner's thought particularly in *Verbal Behavior* (1957), in which he said that any notion of "idea" or "meaning" is explanatory fiction, and that the speaker is merely the locus of verbal behavior, not the cause. Charles Osgood (1957) reinstated meaning in verbal behavior, explaining it as a "representational mediation process" but still did not depart from a generally nonmentalistic view of language. Of further importance to the structural or descriptive linguist was the notion that language could be dismantled into small pieces or units and that these units could be described scientifically, contrasted, and added up again to form the whole. From this principle emerged an unchecked rush of linguists, in the 1940s and 1950s, to the far reaches of the earth to engage in the rigorous production of detailed descriptions of "exotic" languages.

Generative linguistics and cognitive psychology: In the decade of the 1960s, generative-transformational linguistics emerged through the influence of Noam Chomsky and a number of his followers. Chomsky was trying to show that human language cannot be scrutinized simply in terms of observable stimuli and responses or the volumes of raw data gathered by field linguists. The generative linguist was interested not only in describing language (achieving the level of descriptive adequacy) but also in arriving at an explanatory level of adequacy in the study of language, that is, a "principled basis, independent of any particular language, for the selection of the descriptively adequate grammar of each language" (Chomsky, 1964, p. 63). Early seeds of the generative-transformational revolution were planted near the beginning of the twentieth century. Ferdinand de Saussure (1916) claimed that there was a difference between parole (what Skinner "observes," and what Chomsky called performance), on the one

hand, and langue (akin to the concept of competence, or our underlying and unobservable language ability). A few decades later, however, descriptive linguists chose largely to ignore langue and to study parole, as was noted above. The revolution brought about by generative linguistics broke with the descriptivists' preoccupation with performance—the outward manifestation of language—and capitalized on the important distinction between the overtly observable aspects of language and the hidden levels of meaning and thought that give birth to and generate observable linguistic performance. Similarly, cognitive psychologists asserted that meaning, understanding, and knowing were significant data for psychological study. Instead of focusing rather mechanistically on stimulus-response connections, cognitivists tried to discover psychological principles of organization and functioning. David Ausubel (1965, p. 4) noted: from the standpoint of cognitive theorists, the attempt to ignore conscious states or to reduce cognition to mediational processes reflective of implicit behavior not only removes from the field of psychology what is most worth studying but also dangerously oversimplifies highly complex psychological phenomena. Cognitive psychologists, like generative linguists, sought to discover underlying motivations and deeper structures of human behavior by using a rational approach. That is, they freed themselves from the strictly empirical study typical of behaviorists and employed the tools of logic, reason, extrapolation, and inference in order to derive explanations for human behavior. Going beyond merely descriptive adequacy to explanatory power took on utmost importance. Both the structural linguist and the behavioral psychologist were interested in description, in answering what questions about human behavior; objective measurement of behavior in controlled circumstances. The generative linguist and cognitive psychologist were, to be sure, interested in the what question; but they were far more interested in a more ultimate question, why: what underlying factors—innate, psychological, social, or environmental circumstances—caused a particular behavior in a human being? If you were to observe someone walk into your house, pick up a chair and fling it through your window, and then walk out, different kinds of questions could be asked. One set of questions would relate to what happened: the physical description of the person, the time of day, the size of the chair, the impact of the chair, and so forth. Another set of questions would ask why the person did what he or she did: what were the person's motives and psychological state, what might have been the cause of the behavior, and so on. The first set of questions is very rigorous and exacting: it allows no flaw, no mistake in measurement; but does it give you ultimate answers? The second set of questions is richer, but obviously riskier. By daring to ask some difficult questions about the unobserved, we may lose some ground but gain more profound insight about human behavior.

Constructivism- a multidisciplinary approach: Constructivism is hardly a new school of thought. Jean Piaget and Lev Vygotsky, names often associated with constructivism, are not by any means new to the scene of language studies. Yet, in a variety of post-structuralist theoretical positions, constructivism emerged as a prevailing paradigm only in the last part of the twentieth century, and is now almost orthodoxy. A refreshing characteristic of constructivism is its integration of linguistic, psychological, and sociological paradigms, in contrast to the professional chasms that often divided those disciplines in the previous century. Now, with its emphasis on social interaction and the discovery, or construction, of meaning, the three disciplines have much more common ground. What is constructivism, and how does it differ from the other two viewpoints described above? First, it will be helpful to think of two branches of constructivism: cognitive and social. In the cognitive version of constructivism, emphasis is placed on the importance of learners constructing their own representation of reality. "Learners must individually discover and transform complex information if they are to make it their own, [suggesting] a

more active role for students in their own learning than is typical in many classrooms" (Siavin, 2003, pp. 257-258). Such claims are rooted in Piaget's (1954, 1955, 1970; Piaget & Inhelder, 1969) seminal work in the middle of the twentieth century, but have taken that long to become widely accepted views, for Piaget, "learning is a developmental process that involves change, self-generation, and construction, each building on prior learning experiences" (Kaufman, 2004, p. 304). Social constructivism emphasizes the importance of social interaction and cooperative learning in constructing both cognitive and emotional images of reality. Spivacy (1997, p. 24) noted that constructionist research tends to focus on "individuals engaged in social practices, ... on a collaborative group, [or] on a global community." "The champion of social constructivism is Vygotsky (1978), who advocated the view that "children's thinking and meaning-making is socially constructed and emerges out of their social interactions with their environment" (Kaufman, 2004, p. 304).

Topic-215: Nineteen Centuries of Language Teaching

Kelly's (1969) informative survey of language teaching over "twenty-five centuries" revealed interesting anecdotal accounts of foreign language instruction but few if any research-based language teaching methods. In the Western world, "foreign" language learning in schools was synonymous with the learning of Latin or Greek. Latin, though, to promote intellectuality through "mental gymnastics," was until relatively recently held to be indispensable to an adequate higher education. Latin was taught by means of what has been called the Classical Method: focus on grammatical rules, memorization of vocabulary and of various declensions and conjugations, translation of texts, doing written exercises. As other languages began to be taught in educational institutions in the eighteenth and nineteenth centuries, the Classical Method was adopted as the chief means for teaching foreign languages. Little thought was given at the time to teaching oral use of languages; after all, languages were not being taught primarily to learn oral/aural communication, but to learn for the sake of being "scholarly" or, in some instances, for gaining a reading proficiency in a foreign language. Since there was little if any theoretical research on second language acquisition in general, or on the acquisition of reading proficiency, foreign languages were taught as any other skill was taught. So language teaching before the twentieth century is best captured as a "tradition" that, in various manifestations and adaptations, has been practiced in language classrooms worldwide even up to the present time. Late in the nineteenth century, the Classical Method came to be known as the Grammar Translation Method. There was little to distinguish Grammar Translation from what had gone on in foreign language classrooms for centuries, beyond a focus on grammatical rules as the basis for translating from the second to the native language. But the Grammar Translation Method remarkably withstood attempts at the outset of the twentieth century to "reform" language teaching methodology, and to this day it remains a standard methodology for language teaching in educational institutions. Prator and Ceice-Murcia (1979, p. 3) listed the major characteristics of Grammar Translation:

1. Classes taught in the mother tongue; little use of the L2.
2. Much vocabulary taught in the form of lists of isolated words.
3. Elaborate explanations of the intricacies of grammar.
4. Reading of difficult classical texts begun early.
5. Texts treated as exercises in grammatical analysis.
6. Occasional drills and exercises in translating sentences from LI to L2.
7. Little or no attention to pronunciation.

It is remarkable, in one sense, that this method has been so stalwart among many competing models. It does virtually nothing to enhance a student's communicative ability in the language. It is "remembered with distaste by thousands of school learners, for whom foreign language learning meant a tedious experience of memorizing endless lists of unusable grammar rules and vocabulary and attempting to produce perfect translations of stilted or literary prose."

Topic-216: Language Teaching in the Twentieth Century

Against the backdrop of the previous 19 centuries, a glance through the past century or so of language teaching gives us, ironically, a rather refreshingly interesting picture of varied interpretations of the "best" way to teach a foreign language. Perhaps beginning with Francois Gouin's (1880) Series Method, foreign language teaching underwent some revolutionary trends, all of which in one way or another came under the scrutiny of scientific (or observational) research. As schools of thought have come and gone, so have language teaching trends waxed and waned in popularity. Historically, pedagogical innovation has been the beneficiary of the theoretical research described in the previous section, as witnessed by the influence of such research on trends in language teaching. At the same time, language classrooms and their innovative teachers and students have been laboratories of research that have, in turn, informed theoretical stances as they have changed over time. Albert Marckwardt (1972, p. 5) saw these "changing winds and shifting sands" as a cyclical pattern in which a new paradigm (to use Kuhn's term) of teaching methodology emerged about every quarter of a century, with each new method breaking from the old but at the same time taking with it some of the positive aspects of the previous paradigm. More recently, Mitchell and Vidal (2001) described our perhaps misguided penchant for characterizing the last century of language teaching metaphorically as a pendulum swinging back and forth between a number of opposing options: focus on accuracy vs. focus on fluency, separation of skills vs. integration of skills, and teacher-centered vs. learner-centered approaches, to name a few. Mitchell and Vidal suggested that a new metaphor may better depict our journey across time: "that of a major river, constantly flowing, fed by many sources of water—rivers, streams, springs in remote territories, all fed by rain on wide expanses of land" (p. 27). One of the best examples of both the cyclical and fluvial nature of methods is seen in the revolutionary Audiolingual Method (ALM) of the late 1940s and 1950s. The ALM with its overemphasis on oral production drills, borrowed tenets from its predecessor by almost half a century i.e. the Direct Method, but had essentially sprung from behavioral theories of learning of the time. The field of psychology, as noted above in outlining tenets of constructivism, has witnessed a growing interest in interpersonal relationships, the value of group work, and the use of numerous cooperative strategies for attaining desired goal. The same era has seen linguists searching ever more deeply for answers to the nature of communication and communicative competence and for explanations of the interactive, socio-cultural process of language acquisition. The language teaching profession has mirrored these theoretical trends with approaches and techniques that have stressed the importance of self-esteem, intrinsic motivation, students cooperatively learning together, of developing individual strategies for constructing meaning, and above all of focusing on the communicative process in language learning. Some of these methodological innovations will be described in subsequent chapters of this book, as they pertain to issues and topics being discussed. Today, many of the pedagogical springs and rivers of the last few decades are appropriately captured in the term Communicative Language Teaching (CLT), now a catch phrase for language teachers. CLT is an eclectic blend of the contributions of previous methods into the best of what a teacher can provide in authentic uses of the second language in the classroom. Indeed, the single greatest challenge in the profession is to move significantly beyond the teaching of rules,

patterns, definitions, and other knowledge "about" language to the point that we are teaching our students to communicate genuinely, spontaneously, and meaningfully in the second language.

LEARNING STYLE

Topic-217: Learning Style: Field Independence

Learning styles mediate between emotion and cognition, as you will soon discover. For example, a reflective style invariably grows out of a reflective personality or a reflective mood. An impulsive style, on the other hand, usually arises out of an impulsive emotional state. People's styles are determined by the way they internalize their total environment, and since that internalization process is not strictly cognitive, we find that physical, affective, and cognitive domains merge in learning styles. Some would claim that styles are stable traits in adults. This is a questionable view, as noted by Donvyei and Skehan (2003, p.602): "a predisposition may be deep-seated, but it does imply some capacity for flexibility, and scope for adaptation of particular styles to meet the demands of particular circumstances." It would appear that individuals show general tendencies toward one style or another, but that differing contexts will evoke differing styles in the same individual. Perhaps an "intelligent" and "successful" person is one who is "bicognitive"—one who can manipulate both ends of a style continuum.

Field independence: Do you remember, in those coloring books you pored over as a child, a picture of a forest scene with exotic trees and flowers, and a caption saying, "Find the hidden monkeys in the trees"? If you looked carefully, you soon began to spot them, some upside-down, some sideways, some high and some low. A dozen or so monkeys camouflaged by the lines of what at first sight looked like just leaves and trees. The ability to find those hidden monkeys lunged upon your field independent style: your ability to perceive a particular, relevant item or factor in a "field" of distracting items. In general psychological terms, that field may be perceptual, or it may be more abstract and refer to a set of thoughts, ideas, or feelings from which your task is to perceive specific relevant subsets. Field dependence is, conversely, the tendency to be "dependent" on the total field so that the parts embedded within the field are not easily perceived, although that total field is perceived more clearly as a unified whole. Field dependence is synonymous with field sensitivity, a term that may carry a more positive connotation. A field independent (FI) style enables you to distinguish parts from a whole, to concentrate on something (like reading a book in a noisy train station), or to analyze separate variables without the contamination of neighboring variables.

Topic-218: Learning Styles in the Classroom

1. Auditory and musical learners

Auditory learners like to hear solutions and examples explained to them and may gravitate towards music subjects and group learning as a way to understand information. Auditory learners often have a high aptitude for distinguishing notes and tones in music and speech.

Qualities often associated with auditory learners include:

- Possessing a 'good ear' for music and tones
- May be distractible
- Likes to talk to self /others /hum /sing

Auditory learners might say words out loud or hum tones to better learn them. This strategy is a key for keeping musical learners engaged in class lessons.

2. Visual and spatial learner

Visual learners like diagrams, drawing out concepts, charts and processes. They learn by looking at visual concepts, creating them, and watching other people create them. Visual learners might be organized or creative in their application and find things like colors and shapes useful.

Visual learners often possess the following qualities:

- Habitual doodlers / drawers
- Observant
- Not easily distracted
- Enjoys planning
- Prefers visual instructions

3. Verbal learner

Verbal learning includes both writing and speaking. Verbal learners might have a preference for reading and writing, word games, and poems. Verbal learners know the meanings of a broad category of words, can use them effectively, and actively seek out new words to add to their repertoire.

Some qualities associated with verbal learners include:

- Intellectual
- Bookworm
- Good story teller

Verbal learners often seek out careers in journalism and writing, administration, law and politics.

Physical or kinesthetic learner

Commonly called hands-on learners, kinesthetics prefer to physically engage with the materials of the subject matter. Some qualities associated with physical learners include:

- Preference to 'get their hands dirty'
- Energetic, may drum fingers or shake legs
- Action-orientated and outgoing
- May deprioritize reading and writing

Physical learners represent about 5% of the population, and gravitate towards careers with lots of hands on work like emergency services, physical education, and sports.

Social and interpersonal learner

Social learners show preference towards groups and collaboration. Some, but not all, will gravitate towards leadership within a group. Some of the qualities often associated with this type of learner include:

- Extraverted
- Good communicator
- Sensitive and empathetic

It's important for educators to understand that not all social learners are extraverted or highly communicative, and that they can also be visual, auditory, verbal, logical, or physical learners. The interpersonal aspect perhaps better describes the settings in which they are most comfortable, rather than how they absorb information.

As such, teachers should be cognizant of the breadth of variation between different types of social learners. For example, social doesn't strictly mean verbal. Some social learners prefer to listen in a group setting, rather than on their own.

Topic-219: Ambiguity Tolerance

A third style concerns the degree to which you are cognitively willing to tolerate ideas and propositions that run counter to your own belief system or structure of knowledge. Some people are, for example, relatively open-minded in accepting ideologies and events and facts that contradict their own views; they are ambiguity tolerant, that is, more content than others to entertain and even internalize contradictory propositions. Others, more closed-minded and dogmatic, tend to reject items that are contradictory or slightly incongruent with their existing system; in their ambiguity intolerance, they wish to see every proposition fit into an acceptable place in their cognitive organization, and if it does not fit, it is rejected.

Reflectivity and impulsivity

It is common for us to show in our personalities certain tendencies toward reflectivity sometimes and impulsivity at other times. Psychological studies have been conducted to determine the degree to which, in the cognitive domain, a person tends to make either a quick or gambling (Impulsive) guess at an answer to a problem or a slower, more calculated (reflective) decision. David Ewing (1977) referred to two styles that are closely related to the reflectivity/impulsivity (R/I) dimension; systematic and intuitive styles. An intuitive style implies an approach in which a person makes a number of different gambles on the basis of "hunches," with possibly several successive gambles before a solution is achieved. Systematic thinkers tend to weigh all the considerations in a problem, work out all the loopholes, and then, after extensive reflection, venture a solution. The implications for language acquisition are numerous, it has been found that children who are conceptually reflective tend to make fewer errors in reading than impulsive children (Kagan, 1965); however, impulsive persons are usually faster readers, and eventually master the "psycholinguistic guessing game" (Goodman, 1970) of reading so that their impulsive style of reading may not necessarily deter comprehension.

Topic-220: The Role of Learning Styles in the Teaching/Learning Process

Learning styles are individual differences in learning. An individual's learning style "is the way he or she concentrates on, processes, internalizes, and remembers new and difficult academic information or skills" [7, p. 1]. Individuals approach learning differently due to differences in their learning styles. A person's approach to learning is a relatively stable indicator of how they perceive, interact with, and respond to the learning environment.

Understanding learning styles and the role of learning styles in the teaching/learning process is a key component in effective teaching. According to Sarasin, "teaching cannot be successful without knowledge of learning styles and a commitment to matching them with teaching styles and strategies". Utilizing learning style theory in the classroom is extremely beneficial at all educational levels for a variety of reasons. Some research has found a relationship between occupational preferences and learning style type. In addition, student's learning styles have been shown to be affected by their educational experiences, particularly at the postsecondary level. Individuals pursuing careers in information technology (IT) typically encounter a professional work environment that exploits extensive problem solving which draws upon their abilities with a kinesthetic learning style. Therefore, knowledge of learning styles is useful in designing classroom activities that support the development of this learning style.

Topic-221: Visual, Auditory, and Kinesthetic Styles

Yet another dimension of learning style—one that is salient in a formal classroom setting—is the preference that learners show toward either visual, auditory, and/or kinesthetic input. Visual learners tend to prefer reading and studying charts, drawings, and other graphic information. Auditory learners prefer listening to lectures and audiotapes. And kinesthetic learners will show a preference to demonstrations and physical activity involving bodily movement. Of course, most successful learners utilize both visual and auditory input, but slight preferences one way or the other may distinguish one learner from another, an important factor for classroom instruction. In one study of adult learners of ESL, Joy Reid (1987) found some significant cross-cultural differences in visual and auditory styles. By means of a self-reporting questionnaire, the subjects rated their own preferences. The students rated statements like "When I read instructions, I learn them better" and "I learn more when I make drawings as I study" on a five-point scale ranging from "strongly agree" to "strongly disagree." Among Reid's results: Korean students were significantly more visually oriented than native English-speaking Americans; Japanese students were the least auditory students, significantly less auditory inclined than Chinese and Arabic students. Reid also found that some of the preferences of her subjects were a factor of gender, length of time in the United States, academic field of study, and level of education. Later, Reid (1995) reported on studies that included kinesthetic styles with results that confirmed the importance of attending to such preferences among learners. Research findings on learning styles underscore the importance of recognizing learners' varying preferences. However, teachers must take a cautious approach. Measurement of style preferences (usually by means of self-check questionnaires) is problematic (Ehrman & Leaver, 2003).

Topic-222: Students' Diverse Learning Styles in Learning English as A Second Language

In the process of learning the language, there are many variables that determine the success of a language learner. Language learning success is associated with a range of factors including age, gender, motivation, intelligence, anxiety level, learning strategies and language learning styles (Sharp, 2004). In a class made up of various learning styles, it is always necessary for the teachers, particularly the language teachers to identify, respect, and work on the diversity of the learners' differences.

Students may have different levels of motivation, different attitudes about teaching and learning, and different responses to specific classroom environments and instructional practices. The more instructors understand the differences, the better chance they have of meeting the diverse learning needs of their students. This phenomenon was proven true according to the Spolsky's general model of second language learning (1989).

Perceptual learning style is an approach to learning through the five senses. It comprises of auditory learner, visual learner, tactile learner, kinesthetic learner, and haptic learner. Auditory learners learn more through hearing. Visual learners learn more through seeing. Tactile learners discover things through sense of touch. Kinesthetic learners enjoy learning through movement and body experience. Haptic learners are the combination of tactile and kinesthetic learners where they learn more through sense of touch and body involvement. Environmental learning style, on the other hand, comprises of only one dimension which is physical versus sociological. Physical learners can learn better when there are variables such as temperature, sound, light, food, time, and classroom management. These variables have to be taken into considerations during the learning process. In contrast, sociological learners are motivated to learn when there are variables such as group, individual, pair and team work, and level of teacher authority. These variables are important in encouraging the students' motivation to learn.

LEARNING STRATEGIES

Topic-223: Autonomy, Awareness, and Action Strategies

Implied in any consideration of the role of styles and strategies in learning a second language are three linked concepts: autonomy, awareness, and action. These three "As" of learner development have taken on significance in recent years, especially with increasing pedagogical emphasis on learner-centered language teaching (Wenden, 2002). A review of the history of language teaching will reveal some interesting "changing winds and shifting sands." One way of looking at this history is to consider the extent to which methodological trends have emphasized the respective roles of the teacher and the learner.

Topic-224: Learning Strategies

Learning strategies refer to students' self-generated thoughts, feelings, and actions, which are systematically oriented toward attainment of their goals. Metacognitive strategies refers to methods used to help students understand the way they learn; in other words, it means processes designed for students to 'think' about their 'thinking'. Cognitive strategies are more limited to specific learning tasks and involve more direct manipulation of the learning material itself. Socioaffective strategies have to do with social-mediating activity and interacting with others. Learning strategies, as opposed to communication strategies, typically involve the receptive skills of listening and reading. Gender has been shown to be a significant variable in strategy use, both in the case of learning and in communication strategies. Hence, learning strategies help students to self-regulate their learning styles which facilitate them to achieve their goals.

Topic-225: Communication Strategies

In the course of learning a second language, learners will frequently encounter communication problems caused by a lack of linguistic resources. Communication strategies are strategies that learners use to overcome these problems in order to convey their intended meaning. Strategies used may include paraphrasing, substitution, coining new words, switching to the first language, and asking for clarification. These strategies, with the exception of switching languages, are also used by native speakers.

While learning strategies deal with the receptive domain of intake, memory, storage, and recall, communication strategies pertain to the employment of verbal or nonverbal mechanisms for the productive communication of information. In the arena of linguistic interaction, it is sometimes difficult, of course, to distinguish between the two, as Tarone (1983) aptly noted, since comprehension and production can occur almost simultaneously.

Topic-226: Avoidance Strategies

A language learner aims to convey messages to the other people he communicates while producing a foreign or second language and to reach this aim he makes use of various strategies. Corder (1978:18) points out this fact and states that in such a situation the learner has to choose one of the two principal *macro strategies* available to him. In the first case, he may attempt to use all the linguistic

sources at his disposal and pass the precise information to the party he interacts. The learner may paraphrase the message, invent new words, guess words, and borrow some items from his mother tongue.

Topic-227: Compensatory Strategies

Circumlocution: Describing or exemplifying the target object of action (e.g., the thing you open bottles with for corkscrew)

Approximation: Using an alternative term which expresses the meaning of the target lexical item as closely as possible (e.g., ship for sailboat)

Use of all-purpose words: Extending a general, empty lexical item to contexts where specific words are lacking (e.g., the overuse of thing, stuff, what-do-you-call-it, thingie)

Word coinage: Creating a non-existing L2 word based on a supposed rule (e.g., vegeta nan is vegetarian)

Prefabricated patterns: Using memorized stock phrases, usually for "survival" purposes (e.g., Where is the _____ or Comment allez -vous? where the morphological components are not known to the learner)

Nonlinguistic signals: Mime, gesture, facial expression, or sound imitation

Literal translation: Translating literally a lexical item, idiom, compound word, or structure from LI to L2

Foreignizing: Using a LI word by adjusting it to L2 phonology (i.e., with a L2 pronunciation) and/or morphology (e.g., adding to it a L2 suffix)

Code-switching: Using a LI word with LI pronunciation or a L3 word with L3 pronunciation while speaking in L2

Appeal for help: Asking for aid from the interlocutor either directly (e.g., what do you call . . .?) or indirectly (e.g., rising intonation, pause, eye contact, puzzled expression)

Topic-228: Strategy Based Instructions

Much of the work of researchers and teachers on the application of both learning and communication strategies to classroom learning has come to be known generically as **strategies-based instruction (SBI)** (McDonough, 1999; Cohen. 1998), or as learner strategy training. Cohen (1998) likes to refer to "5581"—styles and strategies-based instruction—to emphasize the productive link between styles and strategies. As we seek to make the language classroom an effective milieu for learning, it has become increasingly apparent that "teaching learners how to learn" is crucial. Wenden (1985) was among the first to assert that learner strategies are the key to learner autonomy, and that one of the most important goals of language teaching should be the facilitation of that autonomy.

STRATEGIES-BASED INSTRUCTION

Topic-229: Identifying Learners' Styles and Strategies

A number of options are available for helping learners to identify their own styles, preferences, strengths, and weaknesses. The most common method is a self-check questionnaire in which the learner responds to various questions, usually along a scale of points of agreement and disagreement. Oxford's (1995) Style Analysis Survey and Wintergerst, DeCapua, and Verna's (2002) Learning Styles Indicator offer classic examples of directing learners to identify their own style preferences. A similar questionnaire can be found in Brown's (2002) *Strategies for Success*, a self-help guide for English language learners. The most widely used instrument for learners to identify strategies is Oxford's (1990a) Strategy Inventory for Language Learning (SILL), a questionnaire that has now been tested in many countries and translated into several languages. The SILL's 50 items, divided into six categories, each present a possible strategy (i.e., "I use rhymes to remember new English words.") which responders must indicate on a five-point scale of "never true of me" to "always true of me." The identification of preferred strategies for learners is, in one sense, a logical follow-up to a style's inventory.

Topic-230: Students' Awareness of Learning Styles and Perceptions

Learning strategies are specific combinations or patterns of learning activities used during the learning process. The quality of learning outcomes achieved is dependent to a considerable extent on the learning activities used by learners. These learning strategies can be broadly divided into self-regulated strategy in which the students perform most regulation activities themselves, externally regulated strategy in which the students let their learning process to be regulated by teachers/books or lack of regulation when students are unable to regulate their learning process by themselves and also experience insufficient support from external regulation as provided by teachers and learning environment.

Recent research has made it fairly clear that different students have different LSs. LSPs are significantly different in males and females. Read-write and kinesthetic learners who adopt a deep approach learning strategy perform better academically than do the auditory, visual learners who employ superficial study strategies. Much work has been done on studying the individual learning preferences and how instructional methods can be tailored to cater to the different styles. However, individualization of instructional methods has not been shown to contribute significantly to learn outcomes. Studies have also shown that the most effective learners are able to adapt to the style which the learning situation requires. The teachers can help students to develop strategies for adapting to differing situations, especially when LSs do not fit to a task. Awareness of LSs can create a better learning environment by enabling students to use appropriate strategies.

Topic-231: Incorporating (SB1) into the Language Classroom

Several different manifestations of SB1 can be found in language classes around the world. Through checklists, and other methods discussed above, teachers can become aware of students' tendencies and then offer informal, unplanned advice on beneficial in-class and extra-class strategies. They can essentially be attuned to their role as facilitators of strategic action through tips and pointers and perhaps even anecdotes about "how I learned .., when I was in your shoes." Teachers can also help

students to put the results of a styles questionnaire. Once students have had a chance, with no advance "coaching" to fill out the checklist, you can engage them in any or all of the following: (1) a discussion of why they responded as they did, (2) small-group sharing of feelings underlying their responses, (3) an informal tabulation of how people responded to each item, (4) some advice, from your own experience, on why certain practices may be successful or unsuccessful, or (5) reaching the general consensus that responses in the A and B categories are usually indicative of successful approaches to language learning.

Topic-232: Reconsidering a Strategy-Based Instruction (SBI) to Teaching and Learning Another Language

In the context of second-language acquisition (SLA) theory, there are two different types of learning and communication strategy which can be taught explicitly or implicitly. Explicit learning strategy-instruction, as argued by Chamot (2004), primarily is concerned with development of 'students' awareness of the strategy utilized, teachers' modeling of strategic thinking, students' practice with new strategies, students' self-evaluation of the strategies used, and students 'practice in transferring strategies to new tasks.' Oxford's model (1983), on the other hand, focuses on four areas, namely the use of checklists and/or interviews; the embedding of strategies within L2 learners pedagogy and subsequent implicit practice; the utilization of various compensatory techniques to help students overcome their weaknesses instantly; and the introduction of various strategy textbooks as part of content-centered approach (Brown, 2007).

Topic-233: Stimulating Strategic Action Beyond the Classroom

Finally, it is important to note that style awareness and strategic action are not limited to the classroom. Many so-called successful learners have reached their goals of mastery through their own self-motivated efforts to extend learning well beyond the confines of a classroom. Teachers can help learners to achieve this further step toward autonomy by helping learners to look beyond the classroom and the language course they are in. The ultimate purpose in engaging students in SBI is not simply to complete one language course. Teachers can help learners to see that raising their conscious awareness of styles and strategies aids them in the authentic use of language "out diere."

Topic-234: Language Learning Strategies and its Implications for Second Language Teaching

Research has proved that language learners need instruction in 'how' to use strategies efficiently as a way to improve language learning and performance. There are two common ways to approach language learning strategy instruction: uninformed strategy instruction or direct and integrated instruction. In uninformed strategy instruction, students work through materials and activities designed to elicit the use of specific strategies, but students are not informed of the name, purpose, or value of the specific learning strategy. Direct and integrated instruction (O'Malley & Chamot, 1995, p.153) informs learners of the value and purpose of learning strategies and helps learners to use, identify, and develop learning strategies in a systematic way as they learn the target language.

AFFECTIVE FACTORS IN SECOND LANGUAGE ACQUISITION

Topic-235: Affective Factors in Second Language Acquisition: The Affective Domain

The affective domain: Affect refers to emotion or feeling. The affective domain is the emotional side of human behavior, and it may be juxtaposed to the cognitive side. The development of affective states or feelings involves a variety of personality factors, feelings both about ourselves and about others with whom we come into contact. Benjamin Bloom and Iris colleagues (Krathwohl, Bloom, & Masia, 1964) provided a useful extended definition of the affective domain that is still widely used today. 1. At the first and fundamental level, the development of affectivity begins with receiving. Persons must be aware of the environment surrounding them and be conscious of situations, phenomena, people, objects; be willing to receive—to tolerate a stimulus, not avoid it—and give a stimulus their controlled or selected attention. 2. Next, persons must go beyond receiving to responding, committing themselves in at least some small measure to a phenomenon or a person. Such responding in one dimension may be in acquiescence, but in another higher dimension, the person is willing to respond voluntarily without coercion, and then receives satisfaction from that response. 3. The third level of affectivity involves valuing: placing worth on a thing, a behavior, or a person.

Self-esteem

Self-esteem is probably the most pervasive aspect of any human behavior. It could easily be claimed that no successful cognitive or affective activity can be carried out without some degree of self-esteem, self-confidence, knowledge of yourself, and self-efficacy—belief in your own capabilities to successfully perform that activity. Malinowski (1923) noted that all human beings have a need for phatic communion—defining one self and finding acceptance in expressing that self in relation to valued others. Personality development universally involves the growth of a person's concept of self, acceptance of self, and reflection of self as seen in the interaction between self and others. The following is a well-accepted definition of self-esteem (Coopersmith, 1967, pp. 4-5): by self-esteem, we refer to the evaluation which individuals make and customarily maintain with regard to themselves; it expresses an attitude of approval or disapproval, and indicates the extent to which individuals believe themselves to be capable, significant, successful, and worthy. In short, self-esteem is a personal judgment of worthiness that is expressed in the attitudes that individuals hold toward themselves. It is a subjective experience which the individual conveys to others by verbal reports and other overt expressive behavior.

Topic-236: Attribution Theory and Self-Efficacy

Underlying the issues and questions about the role of self-esteem in language learning are the foundational concepts of attribution and self-efficacy. Based on the seminal work of psychologist Bernard Weiner (1986, 1992, 2000), attribution theory focuses on how people explain the causes of their own successes and failures. Weiner and others (Slavin, 2003; Dornyei, 2001b; Williams & Burden, 1997) describe attribution theory in terms of four explanations for success and/or failure in achieving a personal objective: ability, effort, perceived difficulty of a task, and luck. Two of those four factors are internal to the learner: ability and effort; and two are attributable to external circumstances outside of the learner:

task difficulty and luck. According to Weiner, learners tend to explain, that is, to attribute, their success on a task on these four dimensions. Depending on the individual, a number of causal determinants might be cited.

Topic-237: Willingness to Communicate

A factor related to attribution and self-efficacy, one that has seen a surge of recent interest in the research literature is the extent to which learners display a willingness to communicate as they tackle a second language. Willingness to communicate (WTC) may be defined as "an underlying continuum representing the predisposition toward or away from communicating, given the choice". Or, more simply put, "the intention to initiate communication, given a choice". Emerging from studies and assertions about language learners' willingness to communicate and what we in common lay terms sometimes label as "shyness," researchers have now been examining the extent to which WTC is a factor not just in second language acquisition, but one that may have its roots in a learner's first language communication patterns (MacInryre et al., 2002). In an earlier study on WTC, MacInryre et al. (1998) found that a number of factors appear to contribute to predisposing one learner to seek, and another learner to avoid, second language communication. Noting that a high level of communicative ability does not necessarily correspond with a high WTC, MacInryre et al. proposed a number of cognitive and affective factors that underlie the latter: motivation, personality, intergroup climate, and two levels of self-confidence. The first level resembles what has already been described as situational self-esteem, or "state communicative self-confidence" (MacInryre et al., 1998, p. 547), and the second, an overall global level simply labeled "L2 self-confidence. Both self-confidence factors assume important roles in determining one's willingness to communicate. Other studies of WTC generally confirm its relationship to self-efficacy and self-confidence.

Topic-238: Inhibition Risk Taking

The prominent characteristics of good language learners, according to Rubin and Thompson (1982) were the ability to make intelligent guesses. Impulsivity was also described as a style that could have positive effects on language success. And we have just seen that inhibitions, or building defenses around our egos, can be a detriment. These factors suggest that risk taking is an important characteristic of successful learning of a second language. Learners have to be able to gamble a bit, to be willing to try out hunches about the language and take the risk of being wrong. Beebe described some of the negative ramifications that foster fear of risk taking both in the classroom and in natural settings. In the classroom, these ramifications might include a bad grade in the course, a fail on the exam, a reproach from the teacher, a smirk from a classmate, punishment or embarrassment imposed by oneself.

Topic-239: Anxiety

Intricately intertwined with self-esteem, self-efficacy, inhibition, and risk taking, the construct of anxiety plays a major affective role in second language acquisition. Even though we all know what anxiety is and we all have experienced feelings of anxiousness, anxiety is still not easy to define in a simple sentence. Spielberger defined anxiety as "the subjective feeling of tension, apprehension, nervousness, and worry associated with an arousal of the autonomic nervous system." More simply put, anxiety is associated with feelings of uneasiness, frustration, self-doubt, apprehension, or worry. The research on anxiety suggests that anxiety, like self-esteem, can be experienced at various levels (Horwitz,

2001; Oxford. 1999). At the deepest, or global, level, trait anxiety is a more permanent predisposition to be anxious. Some people are predictably and generally anxious about many things. At a more momentary, or situational level, state anxiety is experienced in relation to some particular event or act. As we learned in the case of self-esteem, then, it is important in a classroom for a teacher to try to determine whether a student's anxiety stems from a more global trait or whether it comes from a particular situation at the moment. Personal factors like trait anxiety, because of its global and somewhat ambiguously denned nature, has not proved to be useful in predicting second language achievement (MacInryre Sc. Gardner, 1991c). However, recent research on language anxiety, as it has come to be known, focuses more specifically on the situational nature of state anxiety.

Empathy

The human being is a social animal, and the chief mechanism for maintaining the bonds of society is language. Some approaches to language teaching fail to accomplish the goal of communicativity in the learner by overlooking the social nature of language. While we tend to recognize the importance of the social aspect of language, we also tend to oversimplify that aspect by not recognizing the complexity of the relation between language and society, or by considering socially oriented problems in language learning as a simple matter of "acculturation." Acculturation is no simple process, and it will become clear in this chapter that the social transactions that the second language learner is called upon to make constitute complex endeavors. Transaction is the process of reaching out beyond the self to others, and language is a major tool used to accomplish that process. A variety of transactional variables may apply to second language learning: imitation, modeling, identification, empathy, extroversion, aggression, styles of communication, and others. Two of these variables, chosen for their relevance to a global understanding of second language acquisition, will be treated here: empathy and extroversion. In common terminology, empathy is the process of "putting yourself into someone else's shoes," of reaching beyond the self to understand what another person is feeling. It is probably the major factor in the harmonious coexistence of individuals in society. Language is one of the primary means of empathizing, but nonverbal communication facilitates the process of empathizing and must not be overlooked. In more sophisticated terms, empathy is usually described as the projection of one's own personality into the personality of others in order to understand them better. Empathy is not synonymous with sympathy. Empathy implies more possibility of detachment; sympathy connotes an agreement or harmony between individuals. Guiora et al. defined empathy as "a process of comprehending in which a temporary fusion of self-object boundaries permits an immediate emotional apprehension of the affective experience of another."

Topic-240: Extroversion

Extroversion and its counterpart, introversion, are also potentially important factors in the acquisition of a second language. The terms are often misunderstood because of a tendency to stereotype extroversion. We are prone to think of an extroverted person as a gregarious, "life of the party" person. Introverts, conversely, are thought of as quiet and reserved, with tendencies toward reclusiveness. Western society values the stereotypical extrovert. Nowhere is this more evident than in the classroom where teachers admire the talkative, outgoing student who participates freely in class discussions. On the other hand, introverts are sometimes thought of as not being as bright as extroverts. Such a view of extroversion is misleading. Extroversion is the extent to which a person has a deep-seated need to receive

ego enhancement, self-esteem, and a sense of wholeness from other people as opposed to receiving that affirmation within oneself. Extroverts actually need other people in order to feel "good." But extroverts are not necessarily loudmouthed and talkative. They may be relatively shy but still need the affirmation of others. Introversion, on the other hand, is the extent to which a person derives a sense of wholeness and fulfillment apart from a reflection of this self from other people. Contrary to our stereotypes, introverts can have an inner strength of character that extroverts do not have. It is unfortunate that these stereotypes have influenced teachers' perceptions of students. Ausubel (1968, p. 413) noted that introversion and extroversion are a "grossly misleading index of social adjustment," and other educators have warned against prejudging students on the basis of perceived extroversion. In language classes, where oral participation is highly valued, it is easy to view active participants with favor and to assume that their visibility in the classroom is due to an extroversion factor (which may not be so). Culturally, American society differs considerably from a number of other societies where it is improper to speak out in the classroom.

MOTIVATION

Topic-241: Theories of Motivation

Various theories of motivation have been proposed over the course of decades of research. Following the historical schools of thought, three different perspectives emerge: 1. from a behavioral perspective, motivation is seen in very matter of fact terms. It is quite simply the anticipation of reward. Driven to acquire positive reinforcement, and driven by previous experiences of reward for behavior, we act accordingly to achieve further reinforcement. Skinner, Pavlov, and Thorndike put motivation at the center of their theories of human behavior. In a behavioral view, performance in tasks—and motivation to do so—is likely to be at the mercy of external forces; parents, teachers, peers, educational requirements, job specifications, and so forth. 2. In cognitive terms, motivation places much more emphasis on the individual's decisions, "the choices people make as to what experiences or goals they will approach or avoid, and the degree of effort they will exert in this respect". Some cognitive psychologists see underlying needs or drives as the compelling force behind our decisions.

Topic-242: Instrumental and Integrative Orientations

Motivation was examined as a factor for a number of different kinds of attitudes. Two different clusters of attitudes divided two basic types of what Gardner and Lambert identified as instrumental and integrative orientations to motivation. The instrumental side of the dichotomy referred to acquiring a language as a means for attaining instrumental goals: furthering a career, reading technical material, translation, and so forth. The integrative side described learners who wished to integrate themselves into the culture of the second language group and become involved in social interchange in that group.

Topic-243: Intrinsic and Extrinsic Motivation

Intrinsically motivated activities are ones for which there is no apparent reward except the activity itself. People seem to engage in the activities for their own sake and not because they lead to an extrinsic reward. Intrinsically motivated behaviors are aimed at bringing about certain internally rewarding consequences, namely, feelings of competence and self-determination. On the other hand, extrinsic motivation is fueled by the anticipation of a reward from outside and beyond the self. Typical extrinsic rewards are money, prizes, grades, and even certain types of positive feedback. Behaviors initiated solely to avoid punishment are also extrinsically motivated. Even though numerous intrinsic benefits can ultimately accrue to those who, instead, view punishment avoidance as a challenge that can build their sense of competence and self-determination.

Topic-244: The Neurobiology of Affect

Using Positron Emission Tomography (PET) and Magnetic Resonance Imaging (MRI) in the empirical study of the brain, some connections have been made between affectivity and mental processing as well as second language acquisition. Research on the neurobiology of affect highlights the roles of the function of Amygdala in language learning and Sustained Deep learning as the key determinants.

Topic-245: Personality Types and Language Acquisition

Within the affective domain, another subarea of interest over the past half century ago has been the measurement of personality characteristics and the hypothesized relationship of such traits to success in various kinds of endeavors. Among dozens of tests and questionnaires designed to tell you more about yourself is the widely-popular Myers-Briggs Type Indicator (Myers, 1962), commonly referred to as the "Myers-Briggs test," borrowing from some of Carl Jung's (1923) "types," the Myers-Briggs team tested four dichotomous styles of functioning in the Myers-Briggs test: (1) introversion vs. extroversion, (2) sensing vs. intuition, (3) thinking vs. feeling, and (4) judging vs. perceiving. There are four categories (Keirsey & Bates, 1984, pp. 25-26) in simple words and self-explanatory phrases. With four two-dimensional categories, 16 personality profiles, or combinations, are possible. Disciples of the Myers-Briggs research (Keirsey & Bates, 1984, for example) described the implications of being an "ENEJ" or an "ISTP," for example, managers may be aided in their understanding of employees by understanding their character type. ISTJs, for example, make better behind-the-scenes workers, while ENFPs might be better at dealing with the public. Lawrence (1984) stressed the importance of a teacher's understanding the individual differences of learners in a classroom: Es will excel in group work; Is will prefer individual work; SJs are "linear learners with a strong need for structure" (p. 52); NTs are good at paper-and-pencil tests.

Topic-246: Measuring Affective Factors

The above discussion of the Myers-Briggs test leads us to probe issues surrounding the measurement of affective factors, which has for many decades posed a perplexing problem. Some affective factors can be reliably measured by means of indirect measures or by formal interviews. But these methods are expensive and require a highly trained expert to administer them. And so, in a spirit of practicality, the language teaching profession has quite consistently relied on "paper-and-pencil" tests, such as the Myers-Briggs, that ask for self-ratings by the learner.

SOCIOCULTURAL FACTORS

Topic-247: Culture: Definitions and Theories

The fact that no society exists without a culture reflects the need for culture to fulfill certain biological and psychological needs in people. Consider the bewildering host of confusing and contradictory facts and propositions and ideas that present themselves every day to anyone; some organization of these facts is necessary to provide some order to potential chaos, and therefore conceptual networks of reality evolve within a group of people for such organization. The mental constructs that enable us thus to survive are a way of life that we call "culture."

Culture establishes for each person a context of cognitive and affective behavior, a template for personal and social existence. But we tend to perceive reality within the context of our own culture, a reality that we have "created," and therefore not necessarily a reality that is empirically defined. "The meaningful universe in which each human being exists is not a universal reality, but 'a category of reality' consisting of selectively organized features considered significant by the society in which he lives" (Condon, 1973, p. 17). Although the opportunities for world travel in the last several decades have increased markedly, there is still a tendency for us to believe that our own reality is the "correct" perception.

Topic-248: Stereotypes or Generalizations

Generalizations become **stereotypes** when all members of a group are categorized as having the same characteristics. **Stereotypes** can be linked to any type of cultural membership, such as nationality, religion, gender, race, or age.

Twain, like all of us at times, expressed caricatures of linguistic and cultural stereotypes. In the bias of our own culture-bound worldview, we too often picture other cultures in an oversimplified manner, lumping cultural differences into exaggerated categories, and then view every person in a culture as possessing stereotypical traits. Here are a few examples: Americans are all rich, informal, materialistic, overly friendly, and drink coffee. Italians are passionate, demonstrative, great lovers, and drink red wine. Germans are stubborn, industrious, methodical, and drink beer. The British are stuffy, polite, thrifty, and drink tea. And Japanese are reserved, unemotional, take a lot of pictures, and also drink tea.

Topic-249: Second Culture Acquisition & Social Distance

Robinson Stuart and Nocon suggested that language learners undergo culture learning as a "process, that is, as a way of perceiving, interpreting, feeling, being in the world, ... and relating to where one is and who one meets" (p. 432). Culture learning is a process of creating shared meaning between cultural representatives. It is experiential, a process that continues over years of language learning, and penetrates deeply into one's patterns of thinking, feeling, and acting.

It is common to describe culture shock as the second of four successive stages of culture acquisition: 1. Stage 1 is a period of excitement and euphoria over the newness of the surroundings. 2.

Stage 2—culture shock—emerges as individuals feel the intrusion of more and more cultural differences into their own images of self and security. In this stage, individuals rely on and seek out the support of their fellow countrymen in the second culture, taking solace in complaining about local customs and conditions, and seeking escape from their predicament. 3. Stage 3 is one of gradual, and at first tentative and vacillating, recovery. This stage is typified by what Larson and Smalley (1972) called "culture stress": some problems of acculturation are solved while other problems continue for some time. But general progress is made, slowly but surely, as individuals begin to accept the differences in thinking and feeling that surround them, slowly becoming more empathic with other persons in the second culture. 4. Stage 4 represents near or full recovery, either assimilation or adaptation, acceptance of the new culture and self-confidence in the "new" person that has developed in this culture.

Topic-250: Teaching Intercultural Competence

While most learners can indeed find positive benefits in cross-cultural living or learning experiences, a number of people experience psychological blocks and other inhibiting effects of the second culture. Stevick (1976b) cautioned that learners can feel alienation in the process of learning a second language, alienation from people in their home culture, the target culture, and from themselves. In teaching an "alien" language, we need to be sensitive to the fragility of students by using techniques that promote cultural understanding. A number of recent research studies have shown the positive effects of incorporating cultural awareness in language classrooms (Byram & Feng, 2005). An excellent set of practical activities, all grounded in research on cultural awareness, is provided in DeCapua and Wintergersts (2004) reference book for teachers. Savignon and Sysoyev (2002) promoted sociocultural competence in their learners of English in Russia by introducing sociocultural strategies such as initiating contact, anticipating cultural misunderstandings, and using diplomacy in discussions. Wright (2000) found that teaching learners of German as a foreign language, using process-oriented tasks promoted cross-cultural adaptability. Abrams (2002) successfully used Internet-based culture portfolios to promote cultural awareness and to defuse cultural stereotypes. Interviews of native speakers of the target language helped learners in Bateman's study (2002) to develop more positive attitudes toward the target culture.

Topic-251: Language Policy, Language Politics, and Word Englishes

Language politics is the way language and linguistic differences between peoples are dealt with in the political arena. This could be manifested as government recognition, as well as how language is treated in official capacities. Some examples: legal status of a language as an official language in a country, state, or other jurisdiction. This generally means that all official documents affecting a country or region are published in the official language(s), but not in those that are not. Evidence in a court of law may also be expected to be presented in an official language. In countries where there is one main language, immigrants seeking full citizenship may be expected to have a degree of fluency in that language ('language politics' then being a reference to the debate over the appropriateness of this). This has been a feature of Australian politics. At various times, minority languages have either been promoted or banned in schools, as politicians have either sought to promote a minority language with a view to strengthening the cultural identity of its speakers, or banning its use (either for teaching, or on occasion an entire ban on its use), with a view to promoting a national identity based on the majority language. An

example of recent promotion of a minority language is Welsh or Leonese by the Leonese City Council; an example of official discouragement of a minority language is Breton.

Language Policy: Many countries have a **language policy** designed to favor or discourage the use of a particular language or set of languages. Although nations historically have used language policies most often to promote one official language at the expense of others, many countries now have policies designed to protect and promote regional and ethnic languages whose viability is threatened. Indeed, whilst the existence of linguistic minorities within their jurisdiction has often been considered to be a potential threat to internal cohesion, states also understand that providing language rights to minorities may be more in their long term interest, as a means of gaining citizens' trust in the central government. Language policy is what a government does either officially through legislation, court decisions, or policy to determine how languages are used, cultivate language skills needed to meet national priorities, or to establish the rights of individuals or groups to use and maintain languages. The scope of language policy varies in practice from state to state. This may be explained by the fact that language policy is often based on contingent historical reasons. Likewise, states also differ as to the degree of explicitness with which they implement a given language policy. The French Toubon law is a good example of explicit language policy. The same may be said for the Charter of the French Language in Quebec.

World Englishes: The rapid growth of English as an international language (EIL) has stimulated interesting but often controversial discussion about the status of English in its varieties of what is now commonly called world Englishes (Kachru, 2005; McKay, 2002; McArthur, 2001; Kachru & Nelson, 1996; Kachru, 1992, 1985). Learning English in India, for example, really does not involve taking on a new culture since one is acquiring Indian English in India. According to Kachru, the "Indianization" of English in India has led to a situation in which English has few if any British cultural attributes. This process of nativization or "indigenization" (Richards, 1979) of English has spread from the inner circle of countries (such as the United States, United Kingdom, Australia, & New Zealand) to an outer circle (Kachru, 1985) of countries that includes India, Singapore, the Philippines, Nigeria, Ghana, and others. In such contexts, English is commonly learned by children at school age and is the medium for most of their primary, secondary, and tertiary education.

Topic-252: ESL and EFL

As the above discussion shows, the spread of EIL has indeed muddled the formerly clear waters that separated what we still refer to as English as a second language (ESL) and English as a foreign language (EFL). Learning ESL—English within a culture where English is spoken natively—may be clearly defined in the case of, say, an Arabic speaker learning English in the United States or the United Kingdom, but not as easily identified where English is already an accepted and widely used language for education, government, or business within the country (for example, learning English in the Philippines or India). According to Nayar (1997), we need to add yet another ESL context; English in Scandinavia, where English has no official status but occupies such a high profile that virtually every educated person can communicate competently with native speakers of English. Learning EFL, that is, English in one's

native culture with few immediate opportunities to use the language within the environment of that culture (for example, a Japanese learning English in Japan), may at first also appear to be easy to define.

NEW DIRECTIONS

Topic-253: New Directions in Language Learning Psychology

Some of the developments that have marked the field in recent years include: broadening the range of constructs investigated, increased recognition of complexity and dynamism, widespread acknowledgement of the situated and social nature of language learner and teacher psychology, and the need for and acceptance of methodological plurality. To meet these new developments, research approaches have shifted from an overwhelming dependence on quantitative designs to the inclusion of more qualitative approaches. Rather than solely testing group averages and correlations, it is now acknowledged as being equally important to qualitatively interpret individual learners' psychology and the contextual factors that interact with this. Achilleas Kostoulas and Juup Stelma who explore the potential of intentionality and Complex Systems Theory (CST) as a basis for a new theoretical direction in language learning psychology, compare two different types of intentionality—a shared 'performance intentionality' in a Norwegian primary English language classroom and a 'competition intentionality' in the context of a private language school in Greece. They demonstrate how intentionalities can orient language learning activity towards particular attractors or preferred states and further argue that intentionalities are linked to differing timescales. Based on these findings, the authors discuss complexity-informed implications for empirical research in language learning psychology.

Topic-254: Mirror Neurons and Language: Challenges and Future Directions

Failure of the mirror neuron system is also now widely accepted as an explanation for congenital neuropsychological deficits, such as autism. Ironically, though, mirror neurons were first discovered in the monkey brain, and monkeys are generally not credited with theory of mind, metaphor, or language—or autism.

The role of mirror neurons in imitation is more contentious. Although mirror neurons have not been recorded directly in humans, brain-imaging studies point to an equivalent system in the human brain, and this system is activated when people imitate action (Nishitani and Hari, 2000, Nishitani and Hari, 2002, Rizzolatti and Craighero, 2004). Yet monkeys appear to be incapable of imitation (Visalberghi and Frigaszy, 1990, Visalberghi and Fragaszy, 2002), suggesting that the mirror system did not evolve to mediate imitation. Rizzolatti and colleagues have suggested instead that the primary role of mirror neurons is in action understanding (Rizzolatti et al., 2001, Rizzolatti and Sinigaglia, 2008); that is, mirror neurons allow the monkey—or human—to understand actions performed by others by mapping those actions onto actions that it can itself perform, but they do not mediate the actual imitation of those actions.

Topic-255: Intentionality and Complex Systems Theory: A New Direction for Language Learning Psychology

Complexity and Complex Systems: Despite increasing interest in CST, there is at present no single authoritative definition of what the theory encompasses. That said, CST, or complexity, can be broadly defined as an ontological and epistemological outlook that is sensitive to the ways in which non-linear, emergent, and holistic phenomena come into existence, without any form of central control, from the interactions of large numbers of entities, or system constituents. A system, in a general sense, is a

collection of entities that exhibit certain behaviors on account of their system membership. For example, students and teachers behave in certain ways because they are part of the school system; celestial bodies in the solar system have certain trajectories because they exert gravitational pull on each other. Complex systems, on the other hand, are more difficult to define, because they are organizationally open. This means that their components interact with, and are influenced by, agents that operate outside the systems' 'boundaries.' Because of the difficulty in separating a system from its surroundings using logical or topographical criteria, it seems preferable to 'frame' systems functionally (i.e., in terms of what the system 'does', or 'what it is for'), while bearing in mind that "the boundary of the system is neither purely a function of our description, nor is it purely a natural thing."

Topic-256: Neurolinguistics Computational Models: Challenges and Future Directions

Module-level structured models: In this section, we will examine work on module-level models. These models attempt to localize processing in particular neural modules. On an anatomical level, it is clear that the brain is rich in structure. For example, there are at least 54 separate processing areas in visual cortex (Van Essen et al. , 1990). But it is not clear whether these areas function as encapsulated modules or rather as interactive pieces of functional networks. Evidence for Neurolinguistics modules has come from three sources: aphasiology, brain imaging, and developmental disorders. The oldest of these sources is the evidence from differing patterns of language deficit in aphasia. One can study patients with lesions of different types in the hope of identifying double disassociations between information-processing skills and lesion types. For example, some patients will have damaged prosodic structure, but normal segmental phonology. Other patients will have damaged segmental structure, but normal prosody. This pattern of results would provide strong support for the notion that there is a localized cognitive module for the processing of prosody. In practice, however, evidence for such double dissociations is difficult to obtain without post hoc partitioning of subject groups. But this partitioning itself casts doubt on the underlying assumptions regarding modularity and dissociability.

Topic-257: New Directions in Language Learning Strategy Research: Engaging with The Complexity of Strategy Use

Definition: Rubin (1975, p. 43) defined learning strategies as, "the techniques or devices which a learner may use to acquire knowledge." Ten years later, although a number of other researchers had contributed to the field (e.g., Stern, 1975; Naiman, Frohlich, Stern, & Todesco, 1978), O'Malley, Chamot, Stewner-Manzanares, Kupper, and Russo (1985) were lamenting the lack of consensus regarding a definition which, they felt, was impeding progress with research. Over the next two decades, however, the controversy continued to rage, until by 2006, Macaro abandoned the attempt to achieve a decisive definition and opted for listing defining characteristics instead. Meanwhile, Dörnyei and Skehan (2003, p. 170) had gone even further and recommended abandoning the term "strategy" in favor of the "more versatile" term "self-regulation." Yet, according to writers such as O'Malley et al. (1985), a definition is necessary for meaningful research. Therefore, following an extensive review of the literature, Griffiths (2008, 2013) distilled a concise definition of language learning strategies, which might be summed up as actions chosen by learners (either deliberately or automatically) for the purpose of learning or regulating the learning of language. This definition depends on the essential features: activity, choice, goal/purpose, and learning.

Topic-258: New Challenges in Psycholinguistics: Interactivity and Alignment in Interpersonal Communication

In recent years, the attention of many psycholinguists has gradually shifted from the individual minds of language users and strictly intra-personal aspects of language processing, towards a different but not necessarily contradictory perspective. Interactivity has come into play as an important factor that can hardly be ignored in the analysis of the mental mechanisms underlying language use. Language emerged from interaction and it has been bound to it during its evolution. Accordingly, to fully capture and understand communicative interactivity, one should take into account that some of its aspects are “embrained” – the neural structures responsible for communication are also interaction-oriented. Moreover, interactivity must also be somehow encoded in the language system itself. In his seminal book, Clark (1996) explicitly states that language should be investigated as a collaborative, joint activity, but even much earlier studies (e.g., Schegloff 1982) refer to discourse as a “an interactional achievement.” A straightforward view of interactivity is that certain actions of one party result in certain actions of the other, which in turn may be noticed by the former and answered with another sequence of actions. Accordingly, one may be able to find some corresponding patterns of modeling in the conversational partners, even though the notion of “correspondence” remains rather intuitive at this stage. The assumed correspondence may be based on synchrony (co-occurrence) or on any kind of similarity relationship between units of communicative modeling. As a result, the activities of conversational parties can be viewed as co-ordinated (Garrod & Anderson 1987). Communicative interactivity is strictly bound to the co-ordination of communicative modeling which can be observed on many levels, from simple physical activities to the dynamics of mental representations, as well as in many domains and modalities. One may assume that there is an inborn tendency to align which may direct humans towards increased interactivity. This tendency may be developed and strengthened due to immersion in society and culture. There are also some external factors, like music, that influence conversational partners and may result in some kind of alignment of their modeling. The present text is intended as a very compact review of various understandings and aspects of interactive alignment. Some issues related to its measurement and modeling are touched upon. Finally, an attempt is made to predict the directions of alignment research and its potential impact on psycholinguistics.

PSYCHOLINGUISTICS IN APPLIED LINGUISTICS: TRENDS AND PERSPECTIVES

Topic-259: Relating Psycholinguistics and Applied Linguistics

If we want to clarify the role psycholinguistics can or should play in AL, we need to narrow down the definition of the latter, or rather look at only a part of that vast field. The acquisition and use of a second language seem to be the appropriate chunk of AL in this context. This sub-area relates to many other parts of our field, but its core is, in my view at any rate, essentially psycholinguistic in nature. The psycholinguistic interest would be in the processing mechanisms involved in using more than one language and the acquisition of additional languages. The AL interest would be in understanding why language learners behave the way they do, or in other words, what the mechanisms are for L2 use and acquisition. Ultimately, interest also lies in interventions that change and improve those mechanisms. This interpretation means that multilingual processing can be defined as the intersection or shared interest across psycholinguistics and AL. In this intersection, there are many questions to be answered: How are different languages processed? What are the processing mechanisms of cross-linguistic influence? What is the impact of level of proficiency? Is there a limit to the number of languages the system can deal with before breaking down? Are there processing differences between different types of languages? To what extent do socio-psychological factors influence processing mechanisms?

Topic-260: Key Issues in The Multilingual Processing: The Structure of The Bilingual Lexicon

For bilingual children who grow up in a bilingual environment, how their language developed through childhood influences the lexical size of both languages. Researchers showed that the basic process is same as with monolinguals, and bilingual children tend to learn the languages as two monolinguals. The growth of both languages' lexicon is the same with the growth of the lexicon for monolinguals. Older children do transfer more than younger children. Also in this step of learning words, the vocabulary size positively related to the exposure time in that language. This will stop until a certain amount of vocabulary of the language is reached. Semantic tasks for preschool children with predominantly Spanish-speaking, predominantly English-speaking, and bilingual showed that these three groups are different from each other. Bilinguals perform best on expressive function for both Spanish and English as predominantly-speaking children but performed differently in each language, which means they do not mirror performance in Spanish and English. They are better on some in English, better on others in Spanish. The ability of learning one language does not influence the ability of learning the other one for bilinguals.

Lexical development in children who learn their second language when their first language is already developed is different from that of children who grow up in a bilingual environment (i.e. simultaneous bilingualism). The beginning step of learning words in the second language is translation, or learning the definitions. This is different from how they learned their first language which involves inputting the information of semantic and formal entities together. When accessing these newly learned words, the basic language semantic system will be activated, which means when a second language word is activated, the basic language word with the same meaning is also activated. It can be

said that learners are still thinking in basic language but try to represent in second language by translation as more semantic and syntax knowledge is learned for the second language. This new language is gradually independent from the basic language. Learners began to access the language without translation with semantic knowledge for that language. As learners gain more and more exposure to the new language, they will complete the development of second language when they can access and use the language from the concept, which can be said to be thinking in that language directly.

Topic-261: Language Choice in Production and Perception

Language choice in production and perception: In both production and perception, there must be language specific processing, though the two processes will differ. In perception, characteristics of the input (e.g., sounds that are language specific) will trigger the system to ‘expect’ input in a given language. Of course, there will also be information in the communicative setting that suggests the use of a particular language. So in perception, language choice is typically both a top-down (setting) and a bottom-up (language characteristics) process. In language production, language choice is essentially a top-down process: the speaker has to include in his/her communicative intention the language in which an utterance has to be encoded. In many situations, it is clear that one specific language has to be selected exclusively for production. In situations in which multilingual speakers are interacting, the use of more than one language is possible and may even be the preferred choice. Language switching can be a communicative tool to highlight specific information or express an attitude towards a topic of conversation. De Bot and Schreuder (1993) propose a ‘language cue’ to explain the wide variation in code switching that has been reported in the literature. They argue that for many individual switches, no linguistic or socio-psychological explanation can be given and indeed is needed because speakers set the cues for the languages to be used to a certain value, leading to the right mix of language in a given situation. The exact locus of the language cue is still a matter of debate.

Topic-262: The Language Mode

In several publications, Grosjean has developed the idea of a language mode to explain the various ways multilingual use their languages. The language mode is defined as follows: “The state of activation of the bilingual’s languages and language processing mechanisms, at a given point in time.” The language mode is a continuum, ranging from a monolingual mode to a bilingual speech mode (Grosjean to appear). In the monolingual mode only one language is activated and the other languages in a multilingual are deactivated. The notion of a language mode is related to the issues of the language cue discussed above: the language mode is defined by the setting and the communicative intentions of a speaker. This is not to say that there is fully conscious control of the position on the mode. Several experimental studies have shown that even in a supposedly monolingual setting, the other language continues to play a role (e.g., in experiments with monolingual and bilingual presentation of stimuli).

Cognitive Processes and SLA

Human cognition is conscious and unconscious, concrete or abstract, as well as intuitive (like knowledge of a language) and conceptual (like a model of a language). It encompasses processes such as memory, association, concept formation, pattern recognition, language, attention, perception, action, problem solving and mental imagery. Traditionally, emotion was not thought of as a cognitive process, but now much research is being undertaken to examine the cognitive psychology of emotion; research is

also focused on one's awareness of one's own strategies and methods of cognition, which is called metacognition.

At one end, the term excludes those children we have just considered who are acquiring two or more languages simultaneously. At the other end, child second-language acquisition generally excludes individuals who are acquiring L2 beyond about 12 years. The reason for this exclusion is that it is commonly thought that there is a critical period for L2 acquisition and that acquiring a second language (or, for that matter, a first one) after puberty is much more difficult. For our purposes, it is generally agreed that child second-language acquisition extends from about 5 to 9 years, or after the primary language is essentially acquired but before any possible effects related to a critical period.

Topic-263: Future Developments and Needs: Language Processing and Language Testing

While language testers are generally never slow or reticent to tell the applied linguistic community what their moral standards and research methods should be, the major part of their work on testing language proficiency is basically built on the black box approach prevalent in the behaviorist era. While there now is quite some information on the various sub processes of language production and language perception, most language testing is still geared toward the outcomes of the whole process. For real diagnostic testing, instruments have to be developed that are specifically aimed at assessing the workings of various sub processes. In production, things can go wrong in many stages of the process. For example, in phonological encoding, segmental and supra-segmental information have to be combined to develop the phonetic plan. It is more or less known how this takes place, and accordingly, what can go wrong. Testing procedures are needed that will allow us to get specific information about problems in these sub stages of language production.

Topic-264: Socio-Psychological Factors in Language Processing

One of the big issues for future research is to determine the extent and the manner in which socio-psychological factors related to the minority status of a language may have an impact on language processing. It is in a way attractive to view our language production system purely as a language producing machine, but this is evidently too simple a picture: lexical access, grammatical complexity, and phonological encoding do not take place in a socio-psychological vacuum. Factors like status, self-esteem, and self-consciousness are critical factors in all stages of the production and perception process. To give an example, when, in speaking, a specific word is needed, there will be a process of matching the meaning components of a lexical item and the communicative intention it is supposed to express. In that matching process, there is an evaluative moment in which many factors will come into play. (For example, is this word appropriate or good enough for this communicative situation? Am I using the right level of politeness? If I cannot use this word, should I continue or stop?) There is no absolute or mechanistic device that can make that decision for all words. Of course, not every single word is weighed in such a way in speaking, since that would lead to too much loss of speed. How such social-psychologically motivated mechanisms operate is far from clear, but, in particular for our understanding of language use in language learners, a better understanding of such mechanisms is vital.

TEACHING TO PSYCHOLINGUISTICS

Topic-265: Writing Systems and Speech

Writing system is based on speech sounds: phonemes or syllables in the sound-based system, each symbol represents a speech sound, either a phoneme or a syllable. There are many different sound-based writing scripts in use throughout the world today – for example, Devanagari in India, Arabic in Egypt, the Hangul syllabary in South Korea, the two Kana syllabaries in Japan, the Cyrillic alphabet in Russia and Bulgaria, and the Roman alphabet in English-speaking countries and Western Europe. Some of these sound-based orthographies correspond highly to their spoken forms. Among these are Finnish and Spanish, which use the Roman alphabet to represent the phonemes of their spoken languages, and Korean and Japanese, which use their own native scripts, Hangul and Kana, respectively, to represent the syllables of their spoken languages. (Complexities occur with Japanese, however, because it also mixes Chinese-type characters into its writing system even though those characters (kanji) can be written in the syllabic forms.) The sound-based orthographies of these languages are easier to read than are sound-based orthographies where the correspondence of written symbol to sound is not high, as is the case for English.

Topic-266: The Study of Writing: Definitions and Classifications

True writing undoubtedly has its roots in pictures for concepts--called pictograms. Pictograms and ideograms are iconic symbols still used in many societies today, both literate and illiterate. Thousands of years ago humans drew pictures of animals and hunters on cave walls. (Lascaux, France) These pictures seem to have been pictograms: that is, pictures representing objects through direct physical resemblance. In Pierce's classification of signs, pictograms are icons, displaying a non-arbitrary relation between form and meaning. Icons, or picture writing used today's industrialized societies include comic strips and political cartoons (minus the captions). It is not known what the pictograms of the Cro Magnon or other ancient groups were used for, but it is certain that they depicted ideas or events and were not a way of symbolizing the sounds of any language. Perhaps we can better guess at the meaning of cave drawings by examining pictographic signs used in more recent times. Many tribal societies have used visual memory aids. This was the function of the Native American wampum belts, which symbolized the main points of treaties, and of the Inc aquipu, or mnemonic knotted strings.

Topic-267: The Whole-Word Vs. Phonics/Decoding Controversy

The Phonics/Decoding approach: The nature of reading according to Phonics/Decoding: The goal of directly determining meaning from written forms is not one shared by Phonics/Decoding advocates. For them, reading is conceived of as a process that converts written forms of language to speech forms and then to meaning (Ehri, 1991; Gough and Juel, 1991; Perfetti, 1991; Adams, 1994; Foorman et al., 1998; National Reading Panel Report, 2000). They regard the essence of reading to be the ability to decode reading materials into speech. Once speech is obtained, they believe, meaning will follow. Thus, they propose early and systematic phonics instruction starting with the mastery of a set of letters and sounds that comprise words.

The Whole-Word approach: Teaching reading should focus on meaning and communication and not on speech. In the view of the Whole-Word proponents, the essential task for a reader is the

recovery of meaning (Huey, 1908; Gates, 1928; Goodman, 1973; F. Smith, 1982, among many others). Whether a reader can say or write the words that are written is incidental to the reading process. Reading is a form of communication the goal of which is the reception of information through written forms.

Fluent Readers Use A Whole-Word Strategy: Actually, all fluent English readers eventually learn to identify whole words as if they were Chinese characters; even the proponents of the Phonics/Decoding Approach admit the fact (see, for example, Byrne, 1992, or Adams, 1994). The time taken to read a page of text aloud is much longer than when the same page is read silently.

Topic-268: Whole Language Instruction Vs. Phonics Instruction

Attitude about phonics instruction: Phonics instruction is the chronological, precise instruction of letter-sound correspondence. It helps students learn to read by recognizing that each letter creates a sound, and when letters are strung together, they make a word (Feng & Maddox, 2013). The five different ways of teaching phonics instruction are analogy phonics, analytic phonics, embedded phonics, phonics through spelling, and synthetic phonics. Analogy phonics helps students connect unknown words to familiar words. Analytic phonics helps students by analyzing letter-sound relationships in order to prevent isolating sounds while reading. Embedded phonics embeds phonic skills into reading text. Phonics through spelling involves selecting letters for phonemes. Synthetic phonics is converting letters to sounds and blending them to make the word sound (National Reading Panel, 2017). When these aspects of phonics are taught systematically with appropriate timing, these methods are successful. In a phonic classroom, the teacher organizes letters and sounds into a sequence (Feng & Maddox, 2013). Phonics instruction claims to teach all children to read by second grade and tends to show quick progress in students. Because the natural alphabetic way, or the phonetic way, is the English language works, supporters of phonics say it allows for strategic learned skills to read any word (Price, 2017). Phonics instruction is not a complete reading program because it lacks comprehension, fluency, and reading strategies.

Topic-269: A Universal Four-Phase Reading Programme

Phase 1: Word Familiarization

The purpose of this phase is to acquaint children with the shapes of written words and to have them become aware that different spoken words of the language have different written manifestations.

Phase 2: Word Identification

In this phase, the child learns which particular written words are associated with which particular spoken words or objects. The difference between this phase and the preceding one is that this one requires the use of long-term memory.

Phase 3: Phrase and Sentence Identification

This phase is similar to that of the preceding word identification one, except that larger linguistic units are dealt with. Its goal is for the child to read the largest basic linguistic unit, the sentence.

Phase 4: Paragraphs, Stories, and Book Reading

The paragraph involves the largest meaningful written linguistic unit. It consists of a sequence of two or more sentences that are related to one another. A sequence of paragraphs can make a story.

Topic-270: The Advantages of Early Reading for Pre-school Age Children

Why is early reading beneficial? There are a number of important advantages of teaching reading to children in their pre-school years: 1. Reading is a source of pleasure for the child. It satisfies and stimulates a child's natural curiosity and, as a source of knowledge, enriches the child. The earlier a child discovers this, the more enriched and more deeply attracted to reading the child will become. 2. Love of reading is established. The warm supportive informal atmosphere of the home or the pre-school provides an excellent situation for learning. In such a situation, a positive attitude towards reading can be established without the difficulties that are often encountered in elementary school. 3. Young children learn quickly and easily. They have a remarkable rote memory learning ability and can easily acquire a multitude of written words. The older the children get and the poorer their memory, the more they require additional exposure and practice. 4. The children grow up to be better learners. They will be able to read faster and with better comprehension than they would if they were to start reading later. In addition to these advantages, there is another important general one. Children who learned to read early would not have to use time in elementary school learning to read. More time therefore could be devoted to the acquiring of other kinds of knowledge. This could have the effect of improving the educational level of children in all areas of knowledge. That being the case, early reading can significantly benefit the whole of society.